



Project Report On

Context Construction for Reliable Reasoning in High-Stakes Information Retrieval using Optimal Transport

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CERTIFICATE

*This is to certify that the project report titled "**Context Construction for Reliable Reasoning in High-Stakes Information Retrieval using Optimal Transport**" is a bonafide record of the work done by **Nia Jobby (U2203157)**, **Richard Sabu (U2203179)**, **Sabharish P V (U2203184)**, and **Sangeetha T Samuel (U2203188)**, submitted to Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2022-2026.*

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Abstract

This project presents an Optimal Transport (OT)-based context retrieval framework for improving information retrieval in reasoning-oriented tasks by reformulating retrieval as a context set construction problem rather than a ranking problem. Conventional retrieval systems rank candidate sentences independently, which often produces redundant evidence and fails to capture the full semantic requirements of complex queries. The proposed system instead focuses on selecting a compact set of sentences that collectively provide sufficient semantic coverage for downstream reasoning tasks.

The framework uses Optimal Transport as a semantic coverage signal, where query and candidate sentence embeddings are represented in a shared semantic space and transport cost is used to measure semantic mismatch. A greedy selection strategy incrementally chooses candidate evidence that minimizes transport cost, thereby prioritizing complementary information and reducing redundancy. This enables retrieval decisions to be guided by semantic sufficiency at the set level rather than by individual relevance scores.

Three variants of the framework are implemented: Fixed-Size OT, which constructs context under a predefined size constraint; Adaptive OT, which dynamically expands context until semantic gain becomes negligible; and Temporal OT, which progressively accumulates evidence for sequential and multi-hop reasoning scenarios. These variants support both fixed-budget and adaptive retrieval while sharing a unified OT-based objective.

The system is evaluated on HotpotQA, FEVER, and SciFact benchmark datasets using set-level metrics such as Hit Ratio, average gold evidence coverage, and full evidence recovery. Experimental results show that OT-based retrieval improves semantic coverage over dense retrieval baselines while using smaller and more efficient context sets, establishing semantic coverage-driven retrieval as a scalable and interpretable alternative for reasoning-focused information retrieval systems.

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List of Abbreviations

LLM	Large Language Model
OT	Optimal Transport
RAG	Retrieval-Augmented Generation
DPR	Dense Passage Retrieval
QA	Question Answering
BEIR	Benchmarking Information Retrieval
BOW	Bag of Words
WMD	Word Mover’s Distance
KG	Knowledge Graph

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Chapter 1

Introduction

This chapter introduces the Optimal Transport based Context Construction framework for reasoning-oriented information retrieval. It describes the limitations of traditional ranking-based retrieval systems and encourages the necessity for set-level semantic coverage. The chapter presents the background of modern retrieval approaches, states the main problem addressed in this project, and establishes the foundation for the proposed solution. Subsequent sections describe the objectives, scope, challenges, assumptions, and relevance of the system.

1.1 Background

Modern information retrieval systems are primarily designed as ranking systems. Given a query, they retrieve candidate documents or sentences and rank them based on estimated relevance. Classical probabilistic models rely on term statistics, while modern dense retrieval approaches represent queries and passages in a shared embedding space and select the top-k most similar items using similarity measures such as cosine similarity. To a large extent, these techniques become the foundation of numerous RAG systems, in which retrieved passages are provided to downstream language models in order to reason and generate answers[3].

Even though ranking-based retrieval is effective in search-oriented querying tasks, it is not entirely oriented towards reasoning-based queries like multi-hop question answering and fact verification. In these tasks, answers depend on combining multiple complementary pieces of evidence rather than retrieving a single highly relevant sentence. Increasing the retrieval depth often introduces redundant information instead of improving semantic coverage.

As a result, context size grows without guaranteeing that all aspects of the query are satisfied. This limitation highlights why the independent ranking approach should be abandoned and consider retrieval as a context set construction problem where collective semantic sufficiency becomes the key goal.

1.2 Problem Definition

The central problem addressed in this project is the mismatch between ranking-based retrieval and the requirements of reasoning tasks. Conventional retrieval systems optimize for individual relevance, selecting top-ranked sentences independently. However, reasoning tasks require a compact set of evidence that is collectively adequate to satisfy all semantic requirements of a query. Simply retrieving the highest-scoring sentences does not ensure that complementary information is included.

This issue becomes more pronounced in multi-hop and compositional reasoning environment. Ranking-based methods tend to have single-evidence bias, where similarity mass is focused on one dominant sentence while additional required evidence is overlooked. Increasing the number of retrieved sentences may reduce the chance of missing evidence, but it also increases redundancy and computational cost without a principled stopping mechanism.

Therefore, the problem is to design a retrieval framework that evaluates relevance at the set level rather than at the individual level. The system must construct a minimal context set that achieves semantic sufficiency, avoids redundancy, and adapts context size according to query complexity. Addressing this problem requires redefining retrieval as a context construction task and introducing an explicit mechanism to measure collective semantic coverage.

1.3 Scope and Motivation

Scope

The scope of this project is to develop and evaluate an OT based framework for context construction in reasoning-oriented information retrieval tasks. The system focuses on selecting compact evidence sets at the sentence level from predefined candidate pools containing gold evidences and hard negatives.

The proposed approach is evaluated under controlled conditions to compare different retrieval strategies. Experiments are conducted on benchmark datasets including HotpotQA [10], FEVER[11], and SciFact[12], representing multi-hop question answering, fact verification, and scientific claim verification tasks. The scope of the work is limited to improving semantic coverage during the retrieval stage and does not involve modifying downstream reasoning or generation models.

Motivation

The motivation for this project arises from the limitations of ranking-based retrieval methods in reasoning tasks. Traditional top-k retrieval approaches optimize the relevance of individual sentences but do not ensure that the selected set of sentences collectively satisfies all semantic aspects of a query. In multi-hop queries, dense retrieval models often prioritize a dominant sentence while failing to retrieve additional supporting evidence required for complete reasoning. Increasing the retrieval depth may improve coverage, but it also introduces larger context sets, redundancy, and inefficiency. Furthermore, ranking-based systems lack a principled mechanism to determine when sufficient evidence has been retrieved. This creates a gap between retrieval effectiveness and the actual requirements of reasoning tasks. Therefore, this work is motivated by the need for a principled and training-free framework that can explicitly measure semantic coverage at the set level using Optimal Transport, enabling compact and coverage-aware context construction.

1.4 Objectives

In order to overcome the shortcomings of ranking-based retrieval and obtain set-level semantic coverage, the following objectives are defined:

- **Reformulate Retrieval as Context Construction** : Restructure the retrieval process as a set-level context construction problem, in which the relevance of an individual sentence is determined collectively rather than independently.
- **Design an Optimal Transport-Based Coverage Framework** : Introduce an Optimal Transport model that quantifies the remaining semantic discrepancy between the query and the constructed context set, allowing explicit evaluation of semantic coverage.

- **Construct a Greedy OT-Guided Context Construction Algorithm :** Implement a greedy selection strategy that sequentially constructs the context set by minimizing transport cost while maximizing marginal semantic gain.
- **Implement Multiple Context Construction Variants :** Develop and evaluate three variants:
 - Fixed-Size OT with a predefined context budget
 - Adaptive OT with coverage-based stopping
 - Temporal OT for progressive multi-hop reasoning
- **Ensure Compact and Efficient Evidence Selection :** Achieve strong semantic coverage while maintaining small context sizes, thereby improving efficiency compared to uncontrolled top- k expansion.

1.5 Challenges

The development and evaluation of the OT-based context construction framework involve several challenges. First, replacing ranking with set-level optimization is a complex computation, as Optimal Transport requires solving a transport minimization problem at each selection step. Although dense similarity-based pruning is used in favor of efficiency, maintaining scalability to larger candidate pools is non-trivial. Second, adaptive stopping must balance coverage and compactness; if stopping occurs too early, one may omit relevant complementary evidence, while excessive expansion leads to redundancy. Third, multi-hop reasoning scenarios require careful combination of complementary sentences without relying on individual similarity scores, which is common with ranking-based systems. Ensuring stable semantic saturation behavior while preserving compact context sets is therefore a key technical issue in this framework.

1.6 Assumptions

The proposed OT-based context construction framework is formulated with well stated structural and implementation assumptions. First, it assumes that the candidate pool of sentences for each query is finite and predefined, consisting of gold evidence sentences and hard negatives. This assumption is necessary to ensure that greedy selection operates over a constrained search space and that convergence properties can be proved.

Second, the framework assumes that sentence and query embeddings are generated with the help of a frozen encoder, specifically BAAI/bge-base-en-v1.5. All embeddings are ℓ_2 norm-normalized to unit length. There is no fine-tuning or task-specific adaptation done. This guarantees that comparisons between retrieval methods are made on the same semantic representations.[1]

Finally, the theoretical properties of the system are based on assumptions about the Optimal Transport formulation. The entropically regularized OT functional has the assumption that it is non-negative, and greedy selection guarantees the monotonic decrease of the transport cost at every step. With such assumptions semantic saturation and convergence behavior is ensured in both adaptive and temporal versions of the framework.

1.7 Societal / Industrial Relevance

The proposed OT-based context construction framework is particularly relevant for reasoning-focused information retrieval systems used in knowledge-intensive domains. Applications that involve fact verification, scientific claim validation, and multi-hop question answering require compact and semantically sufficient evidence sets rather than long ranked lists of individually relevant sentences. By explicitly modeling semantic coverage at the set level, the framework addresses the drawbacks of ranking-based retrieval that may overlook complementary evidence needed to make reliable reasoning.

From an industrial perspective, it is significant to control context size in downstream language models which operate under input token constraints. The adaptive OT variant scales context size according to query complexity, while maintaining compact sets compared to uncontrolled top-k expansion. This enhances efficiency without losing semantic coverage.

The temporal OT extension assists in progressive evidence aggregation in multi-hop reasoning scenarios, which is valuable for systems that require sequential reasoning.

From a research perspective, the framework provides a unified, training-free approach to context construction using Optimal Transport as an explicit coverage signal. By reformulating retrieval as a set-level optimization problem, it provides a principled alternative to ranking-based retrieval for reasoning-oriented tasks.

1.8 Organization of the Report

This report is structured as follows:

- **Chapter 1: Introduction** – Presents the background of ranking-based retrieval, defines the problem of set-level semantic coverage, and outlines the objectives, scope, challenges, assumptions, and relevance of the proposed framework.
- **Chapter 2: Literature Survey** – Reviews prior work in ranking-based retrieval, adaptive retrieval depth, diversification methods, set-based optimization, and Optimal Transport approaches relevant to context construction.
- **Chapter 3: Proposed System and Experimental Setup** – Describes the OT-guided context construction framework, including fixed-size, adaptive, and temporal variants, along with datasets, baselines, evaluation protocols, and implementation details.
- **Chapter 4: Results and Conclusion** – Presents experimental results, qualitative analysis, ablation studies, and summarizes the contributions and findings of the proposed system.

This chapter introduced the motivation for redefining retrieval as a context construction problem and established the foundational structure for the remainder of the report.

1.9 Summary of Chapter

This chapter introduced the Optimal Transport-based context construction framework and highlighted the limitations of traditional ranking-based retrieval for reasoning-oriented tasks. It described the necessity to consider semantic sufficiency at the set level rather than relying on independent relevance scores. The issue of single-evidence bias, redundancy, and uncontrolled context growth was defined, motivating the reformulation of retrieval as a context construction task.

The chapter also outlined the scope, objectives, challenges, assumptions, and societal relevance of the proposed framework. By introducing an explicit semantic coverage formulation using Optimal Transport, the project establishes a principled way of constructing compact and sufficient evidence sets for multi-hop question answering and fact verification tasks. The next chapter reviews related work that forms the foundation for this approach.

Chapter 2

Literature Review

2.1 Introduction

The development of open-domain question answering systems has increased interest in effective retrieval strategies for identifying relevant information from large document collections. Traditional methods such as BM25 rely on lexical matching, while recent work has shifted toward dense retrieval techniques that represent queries and passages in continuous embedding spaces. Dense Passage Retrieval (DPR) demonstrated that dual-encoder architectures can significantly improve retrieval accuracy in open-domain QA tasks. Subsequent research has explored methods such as contrastive pre-training and retrieval-augmented generation to further enhance retrieval performance and integrate retrieval with answer generation. The following sections review these approaches and analyze their contributions and limitations in the context of open-domain question answering.

2.2 Existing Techniques

2.2.1 Dense Passage Retrieval (DPR) – Karpukhin et al., 2020 [1]

Problem Addressed

Open-domain question answering involves retrieving relevant passages from a large corpus before extracting the final answer. Traditional sparse retrieval methods such as BM25 rely on lexical overlap, which prevents them to be able to detect deeper semantic relationships between questions and passages. This often results in missing relevant documents when wording differs significantly between the query and the source text. The authors identify the need for a retrieval model that can learn semantic similarity beyond exact term matching.

Objective

The objective of DPR is to develop a dense retrieval framework that learns to represent questions and passages in a shared embedding space. It aims to enhance the accuracy of passage retrieval in open-domain QA by training the model in supervised pairs that consist of question-answer. The retrieved passages should provide strong evidence for downstream reading models to extract correct answers.

Methodology

DPR has a dual-encoder (bi-encoder) design consisting of two separate encoders: one for questions and one for passages. Each encoder converts its input into a fixed-dimensional vector representation. The relevance score between a question and a passage is computed using the inner product of their embeddings.

During training, DPR uses supervised data where each question is paired with a positive passage containing the correct answer. In-batch negatives are used to improve training efficiency, meaning that other passages within the same batch act as negative examples. The model is optimized using a contrastive loss that encourages higher similarity for positive pairs and lower similarity for negative pairs.

At inference time, all passages are encoded offline and indexed. Given a new question, its embedding is computed and compared against passage embeddings to retrieve the top-k most similar passages

Experimental Results

The authors evaluate DPR using open-domain QA benchmarks and found that it achieved better results in passage retrieval accuracy compared to BM25. The model has been shown to perform well with the downstream reading components and it indicates that dense semantic representations can be useful in large scale retrieval tasks.

Strengths

DPR presents an efficient and scalable dense retrieval framework. The use of in-batch negatives improves training efficiency, and the dual-encoder structure allows pre-computation of passage embeddings for fast retrieval. The model demonstrates clear improvements over traditional sparse retrieval in open-domain QA procedures.

Limitations

Regardless of its advancements, DPR relies on supervised training data, which requires large annotated datasets. The retrieval process remains ranking-based and selects a fixed number of top-k passages without directly modeling collective semantic sufficiency. Additionally, performance may degrade when applied to domains that differ from the training distribution.

2.2.2 Relevance-Aware Contrastive Pre-Training (ReContriever) – Lei et al., 2023 [2]

Problem Addressed

Large annotated datasets are effective in training dense retrievers to perform well. However, collecting human-labeled question–passage pairs is expensive and domain-specific. Unsupervised contrastive pre-training approaches like Contriever attempt to overcome this weakness by generating pseudo-positive pairs by randomly cropping text spans.

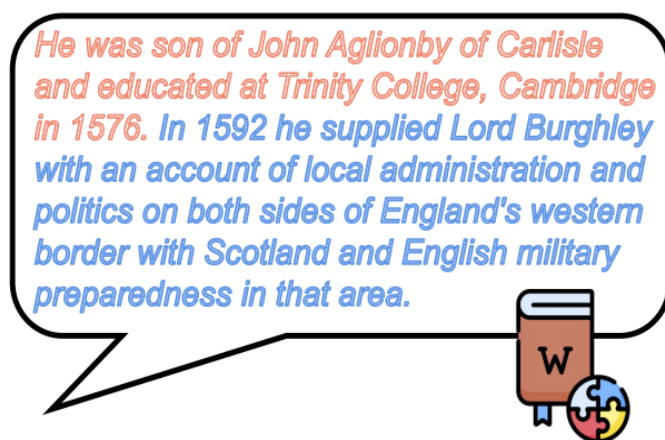


Figure 2.1: A text snippet from Wikipedia, where two nearby sentences are quite irrelevant. Random cropping may lead to a false positive query–passage pair.

The main problem that the authors mention regarding this approach is that randomly sampled positive pairs can be irrelevant in their meaning, resulting in the false positives during the training. These noisy pairs may affect the quality of representation and decrease the retrieval performance.

Objective

The objective of ReContriever is to improve unsupervised dense retrieval by reducing the effects of false positive pairs in contrastive learning. The model is designed to boost the retrieval performance without using human-annotations. It seeks to produce more reliable semantic representations that generalize well across domains and tasks.

Methodology

ReContriever is based on the Contriever framework and introduces two significant changes. First, it takes an one-document-multi-pair strategy, where multiple positive query–passage pairs are generated from the same document instead of only one. This allows better utilization of information within each document.

Second, the model introduces a relevance-aware contrastive loss. During training, the model itself is used as an intermediate oracle to estimate the relevance of each positive pair. These estimated relevance scores are then used to assign adaptive weights to the contrastive loss. Pairs that are more likely to be truly relevant receive higher weights, while less relevant pairs receive lower weights.

The model is evaluated on the BEIR benchmark and open-domain QA retrieval datasets. All training is conducted without human-annotated supervision.

Experimental Results

It has been demonstrated through experiment that ReContriever consistently improves over the original Contriever model across multiple BEIR datasets. It also shows competitive performance in open-domain QA retrieval benchmarks. The model reduces the performance gap between unsupervised dense retrieval and traditional sparse methods such as BM25.

Strengths

ReContriever minimizes the negative impact of noisy pseudo-positive samples in unsupervised training. It improves retrieval performance without any labeled data and shows better generalization across domains. The method remains consistent with standard dense retrieval architectures.

Limitations

Although ReContriever can help to increase the quality of representations, it still follows a ranking-based retrieval paradigm and retrieves a fixed number of top-k passages. The method does not explicitly model collective semantic coverage or adaptive context selection. Additionally, performance remains dependent on the quality of pseudo-positive generation during pre-training.

2.2.3 Retrieval-Augmented Generation (RAG) – Lewis et al., 2020 [3]

Problems Addressed

Large pre-trained language models can represent factual knowledge in their parameters and perform well when fine-tuned. However, they cannot easily update stored knowledge, provide clear provenance for predictions, or reliably access specific external information when needed. In knowledge-intensive tasks such as open-domain question answering and fact verification, purely parametric models may produce incorrect or hallucinated outputs. The authors identify the need for a model that combines parametric memory with explicit non-parametric memory.

Objectives

The objective of RAG is to integrate a neural retriever with a pre-trained sequence-to-sequence generator in a unified framework. The model aims to leverage both parametric memory (stored in model weights) and non-parametric memory (stored in an external document index). This allows the system to retrieve relevant documents and condition generation on them, enhancing factual grounding and adaptability.

Methodology

RAG consists of two main components: a dense retriever and a generator. The retriever is based on Dense Passage Retrieval (DPR) and retrieves the top-K relevant documents using Maximum Inner Product Search . The generator is a pre-trained BART sequence-to-sequence model. Retrieved documents are treated as latent variables, and the model marginalizes over them during the production of the output sequence.

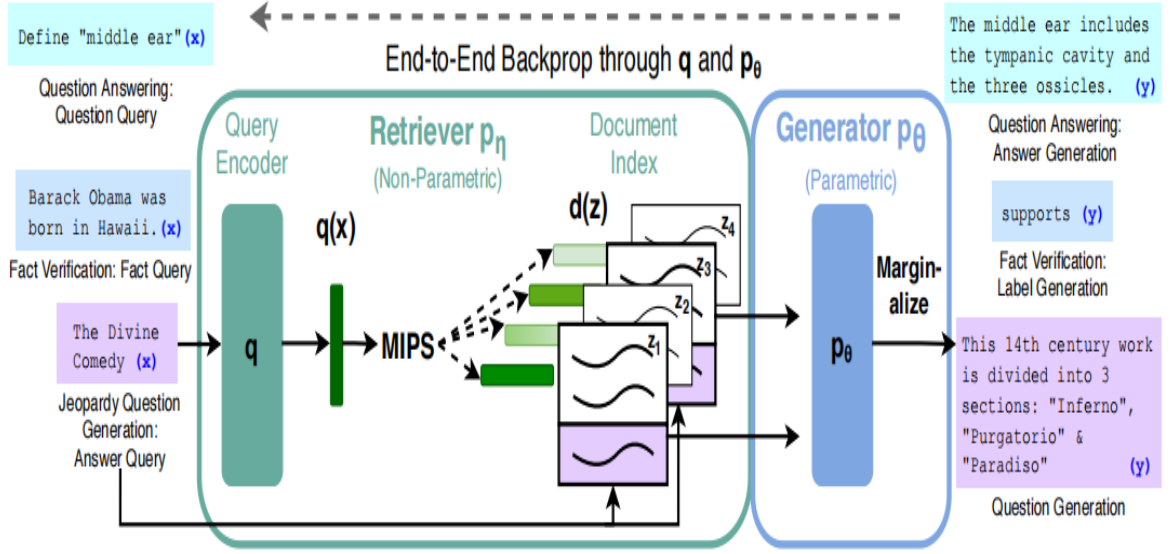


Figure 2.2: Overview of RAG approach

Two variants are proposed. In RAG-Sequence, a single retrieved document is used for generating the entire output sequence. In RAG-Token, different documents can be used for generating different tokens in the sequence. The retriever and generator are trained jointly by minimizing the negative log-likelihood of the target output.

Experimental Results

The authors evaluate RAG on multiple knowledge-intensive tasks including open-domain QA, abstractive QA, question generation, and fact verification. RAG obtains state-of-the-art results on Natural Questions, WebQuestions, and CuratedTrec benchmarks. It also demonstrates improved factuality and specificity in generation tasks compared to parametric-only baselines.

Strengths

RAG is efficient in applying retrieval and generation in the same framework. It allows updating knowledge by replacing the document index without retraining the entire model. The model shows strong performance across multiple tasks and improves factual grounding compared to parametric-only models.

Limitations

RAG is dependent on the quality of the retriever and still retrieves a fixed number of top-K documents. Increasing the number of retrieved documents increases computational cost. Additionally, training requires large memory resources due to the document index and joint optimization process.

2.2.4 Retrieve What You Need – Wang et al., 2024 [4]

Problems Addressed

Retrieve-then-read frameworks typically retrieve a fixed number of passages before answer generation. However, previous studies have shown that using a large number of passages significantly increases memory usage and inference cost, while providing slight improvement in accuracy. The authors observe that many retrieved passages are redundant and unnecessary for answering a given question. This motivates the need to reduce the number of support passages without compromising performance.

Objectives

The objective of this work is to improve both the inference efficiency and predictive performance of retrieve-then-read systems. The authors propose introducing an intermediate module called a knowledge selector that refines the retrieved passage set before it is passed to the reader. The goal is to select a smaller subset of relevant passages while maintaining or improving question answering accuracy.

Methodology

The proposed framework inserts a knowledge selector between the retriever and the reader. The retriever first retrieves a fixed number of passages from a large knowledge source. The knowledge selector then prunes this set to obtain a smaller subset, which is passed to the reader for answer generation.

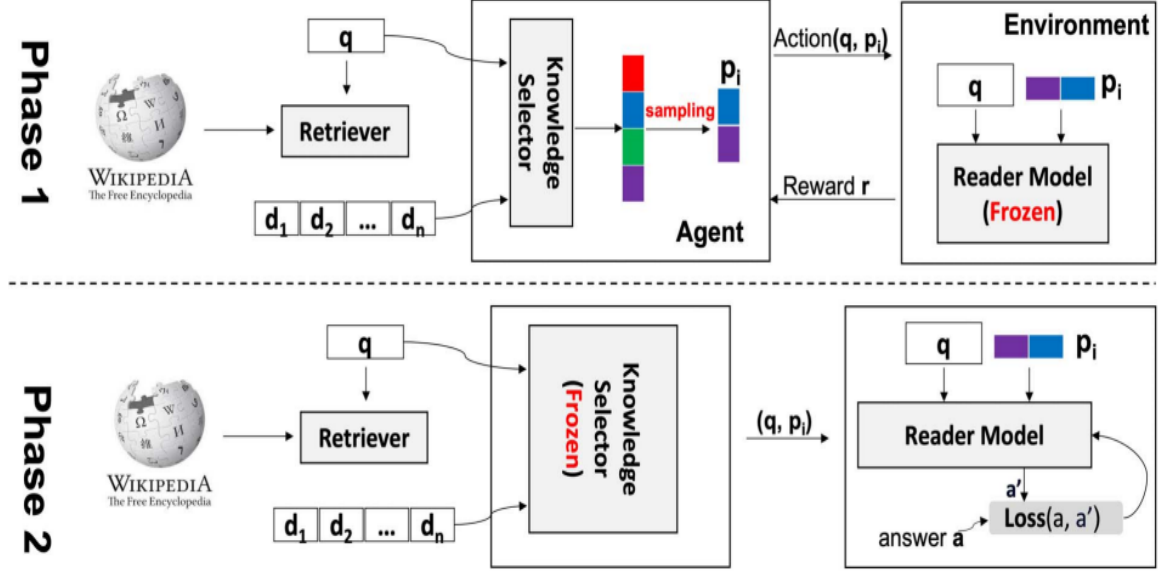


Figure 2.3: Architecture of our proposed mutual learning framework.

The knowledge selector is trained using reinforcement learning, specifically a policy gradient method. It is formulated as a contextual multi-arm bandit problem, where the selector chooses K passages to maximize the reward based on the reader’s answer correctness. A two-phase training strategy is used. In Phase 1, the knowledge selector is trained while keeping the reader fixed. In Phase 2, the reader is trained using the passages selected by the updated selector.

The retriever used is DPR, and the reader is based on the Fusion-in-Decoder (FiD) architecture. Experiments are conducted on Natural Questions, TriviaQA, and WebQuestions datasets.

Algorithm 1 Two-phase Training

```
1: Input:  $D$  : question-answer pairs,  $\mathcal{E}$  : external source,  $epochs$  : number of epochs,  $\phi$   
   : fixed-parameter retriever, initialized knowledge selector  $\pi_\theta$ ,  $n$  : number of passages  
   retrieved by  $\phi$ ,  $K$  : number of passages selected by  $\pi$   
2: for  $e = 1$  to  $epochs$  do  
3:   Phase 1: Train knowledge selector  
4:   for each question  $(q_i, a_i) \in D$  do  
5:     Retrieve  $n$  passages from  $\mathcal{E}$  via  $\phi$   
6:     Select  $K$  passages  $p_i$  out of the  $n$  retrieved passages by  $\pi_\theta(p_i|q_i)$   
7:     Generate  $\hat{a}_i$  by  $\Psi_\phi(a_i|q_i, p_i)$   
8:     Compute gradient of  $\pi_\theta$ :  $r(a_i|q_i, p_i)\nabla_\theta \log \pi_\theta(p_i|q_i)$   
9:     Update the parameters of  $\pi_\theta$   
10:  end for  
11:  Phase 2: Train FiD-based reader  
12:  for each question  $(q_i, a_i) \in D$  do  
13:    Retrieve  $n$  passages from  $\mathcal{E}$  via  $\phi$   
14:    Select  $K$  passages  $p_i$  out of the  $n$  retrieved passages by  $\pi_\theta(p_i|q_i)$   
15:    Generate  $\hat{a}_i$  by  $\Psi_\phi(a_i|q_i, p_i)$   
16:    Compute gradient of  $\Psi_\phi$ :  $\nabla_\phi \mathcal{L}(a_i|q_i, p_i)$   
17:    Update the parameters of  $\Psi_\phi$   
18:  end for  
19:  Save the optimal parameters of both  $\pi_\theta$  and  $\Psi_\phi$  by evaluating the validation dataset  
20: end for
```

Experimental Results

Experimental results indicate that the proposed framework improves top-k retrieval accuracy and end-to-end exact match performance compared to baseline retrieve-then-read systems. The model achieves competitive or state-of-the-art results on NQ, TQA, and WebQ benchmarks while using significantly fewer passages. The authors also report improvements in computational efficiency measured using FLOPs.

Strengths

The framework reduces the number of required support passages while maintaining strong performance. It does not require additional annotated question–passage pairs and can be applied to different retrievers and readers. The two-phase mutual learning strategy improves both retrieval quality and answer generation performance

Limitations

The framework includes additional training complexity due to reinforcement learning and two-phase optimization. It relies on an initial retriever to generate candidate passages and does not eliminate the ranking-based retrieval step entirely. Performance depends on the quality of both the retriever and reader components

2.2.5 From Word Embeddings to Document Distances – Kusner et al., 2015 [5]

Problem Addressed

Traditional document similarity measures such as Bag-of-Words (BOW) and TF-IDF represent documents based on word frequency but do not capture semantic relationships between words. Two documents may convey similar meaning while sharing no common words, resulting in large distance under BOW-based representations. The authors highlight that such representations fail to incorporate semantic similarity between related words such as “Obama” and “President” or “Illinois” and “Chicago”

Objectives

This work aims to characterize a document distance metric that incorporates semantic relationships learned through word embeddings. The authors propose the Word Mover’s Distance (WMD), which measures the minimum cumulative cost required to transform the word distribution of one document into another using distances between embedded word vectors

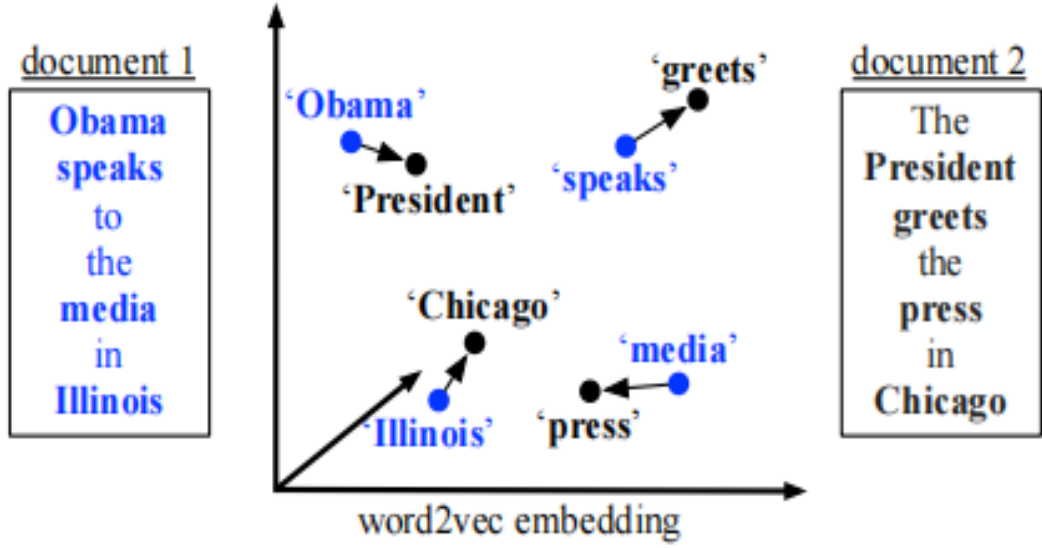


Figure 2.4: All non-stop words (bold) of both documents are embedded into a word2vec space. The distance between the two documents is the minimum cumulative distance that all words in document 1 need to travel to exactly match document 2.

Methodology

The suggested approach represents each document as a normalized bag-of-words (nBOW) vector. Word embeddings are obtained using the word2vec model, where each word is mapped to a dense vector representation in a continuous space.

The distance between two documents is formulated as an optimization problem based on the Earth Mover’s Distance (EMD). A flow matrix is computed that determines how much “mass” from one word in the first document must be transported to a word in the second document. The objective is to minimize the total transportation cost, where the cost between words is defined as the Euclidean distance between their embeddings.

This formulation results in a linear programming problem. The authors also introduce lower bounds such as Word Centroid Distance (WCD) and Relaxed Word Mover’s Distance (RWMD) to accelerate computation.

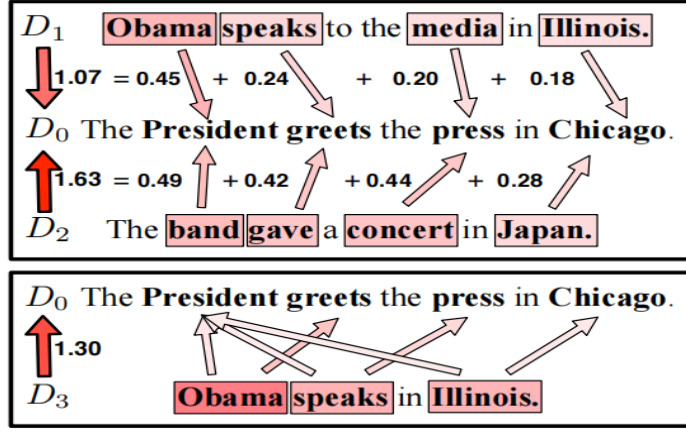


Figure 2.5: Architecture of our proposed mutual learning framework.

Experimental Results

The authors evaluate WMD on eight document classification datasets using k-nearest neighbor classification. Results show that WMD outperforms traditional methods such as BOW, TF-IDF, BM25, LSI, LDA, and other neural representations on most datasets.

The study also demonstrates that WMD benefits from high-quality word embeddings trained on large corpora and achieves lower classification error rates compared to competing document distance metrics.

Strengths

WMD uses word embedding semantic information directly as an input to document distance computation. It is hyperparameter-free and interpretable, as document distances can be explained through word-level transport costs. The method consistently achieves strong performance across multiple classification benchmarks.

Limitations

The primary limitation of WMD is computational complexity, as solving the optimal transport problem is expensive for large documents. Although lower bounds such as RWMD reduce computation time, the method remains slower than simpler distance measures like BOW or TF-IDF.

Table 2.1: Comparison of key related works in semantic retrieval and retrieval-augmented context construction.

Paper Name	Key Concept	Performance	Strengths	Limitations
Dense Passage Retrieval (DPR) [1]	Dual-encoder dense retrieval using question and passage encoders trained with positive and hard negative pairs	Outperformed BM25 by 9–19% absolute top-20 retrieval accuracy	Strong dense retrieval baseline, scalable with FAISS indexing, effective semantic matching	Retrieval optimized for individual passage relevance rather than complementary evidence selection
ReContriever[2]	Enhances unsupervised dense retrieval through relevance-aware contrastive pre-training; adaptively weights pseudo-positive pairs	Improved Contriever on BEIR and open-domain QA retrieval benchmarks	Reduces false positives in unsupervised retrieval; effective with limited supervision	Still dependent on contrastive pair quality; does not explicitly model set-level coverage
Retrieval-Augmented Generation (RAG) [3]	Combines dense retrieval with seq2seq generation using parametric and non-parametric memory; retrieved documents treated as latent variables	Achieved state-of-the-art on multiple open-domain QA benchmarks	End-to-end trainable retrieval-generation framework; supports knowledge updates through external memory	Retrieval remains ranking-based; retrieved passages may still contain redundant context
Retrieve What You Need [4]	Introduces a reinforcement-learning-based knowledge selector between retriever and reader to prune redundant passages in open-domain QA	Improved retrieval quality and QA accuracy while reducing passage count	Reduces redundancy, improves inference efficiency, does not require additional passage annotations	Designed for retrieve-then-read QA pipelines; depends on reader feedback during training
Word Mover’s Distance (WMD) [5]	Measures semantic distance between documents using Earth Mover’s Distance over word embeddings; computes minimum transport cost between word distributions	Outperformed seven document distance baselines on 6 of 8 classification datasets	Hyperparameter-free, interpretable transport formulation, captures semantic similarity beyond lexical overlap	High computational cost for long documents; operates at word level without contextual embeddings

2.2.6 Gaps Identified

From the comparison of the reviewed works, several common patterns emerge. Most retrieval approaches such as DPR and ReContriever treat retrieval as an independent ranking problem where passages are scored individually and the top-k results are selected without ensuring that the chosen set collectively covers all semantic aspects of the query. RAG integrates retrieval with generation, but it still relies on a fixed top-k retrieval stage and does not guarantee that the retrieved documents are jointly sufficient or non-redundant. Retrieve What You Need reduces redundancy using a reinforcement-learning based knowledge selector, yet it still depends on an initial ranking step and lacks an explicit set-level coverage objective. Word Mover’s Distance models semantic similarity using optimal transport, but it operates at a document-to-document level and is computationally expensive for large-scale retrieval. Overall, while existing methods improve retrieval accuracy and integration with generation, they do not explicitly model collective semantic coverage, highlighting the need for approaches that better support reasoning-oriented retrieval tasks.

2.2.7 Conclusion

This chapter reviewed five key works related to dense retrieval, retrieval-augmented generation, adaptive passage selection, and semantic distance modeling. DPR demonstrated the effectiveness of supervised dense retrieval for open-domain question answering, while ReContriever extended dense retrieval to unsupervised settings using contrastive learning. RAG integrated retrieval with sequence-to-sequence generation for knowledge-intensive tasks, and Retrieve What You Need introduced reinforcement learning for adaptive passage selection. Word Mover’s Distance applied optimal transport to capture semantic similarity between documents. Despite these advances, most approaches rely on ranking-based retrieval or document-level similarity and do not explicitly model semantic sufficiency at the set level. This motivates the need for alternative retrieval formulations that treat context construction as a collective optimization problem, which is addressed in the next chapter.

Chapter 3

Methodology

3.1 Problem Reformulation

Conventional information retrieval systems are primarily designed as *ranking systems*. Given a query, these systems retrieve candidate documents or sentences and rank them based on their estimated relevance. However, reasoning-intensive tasks such as multi-hop question answering and fact verification require more than ranked results. They require a compact *evidence set* that collectively captures all the semantic components of the query.

In this project, retrieval is reformulated as the problem of **constructing a context set** that sufficiently represents the semantics of the query.

Let:

- q denote the query,
- $S = \{s_1, s_2, \dots, s_N\}$ denote the pool of candidate evidence sentences,
- $C \subseteq S$ denote the constructed context set.

The objective is to construct a subset C such that:

- The semantic components of the query are adequately represented.
- Additional evidence elements are included only if they contribute meaningful semantic coverage.

Thus, the retrieval task is reformulated from *ranking-based selection* to a **set-level semantic optimization problem**.

3.2 Semantic Representation

To enable semantic comparison, the query and candidate sentences are mapped into a shared vector embedding space.

$$q \in \mathbb{R}^d \quad s_j \in \mathbb{R}^d \quad (3.1)$$

where q represents the query embedding and s_j represents the embedding of the j -th candidate sentence.

For consistency in similarity computation, all embeddings are ℓ_2 normalized:

$$\|q\|_2 = 1 \quad \|s_j\|_2 = 1 \quad (3.2)$$

Semantic similarity between the query and a candidate sentence is computed using cosine similarity:

$$\cos(q, s_j) = q^\top s_j \quad (3.3)$$

Semantic distance is defined as:

$$d(q, s_j) = 1 - \cos(q, s_j) \quad (3.4)$$

3.3 Optimal Transport

Optimal Transport is used to model how the semantic mass of the query is distributed across the selected context elements.

3.3.1 Cost Matrix

Consider a constructed context set:

$$C = \{s_{j_1}, s_{j_2}, \dots, s_{j_K}\} \quad (3.5)$$

The cost between the query and each context element is defined as:

$$C_{1,k} = 1 - \cos(q, s_{j_k}) \quad (3.6)$$

3.3.2 Marginal Distributions

The source and target distributions are defined as uniform:

$$a = [1], \quad b = \frac{1}{K} \mathbf{1}_K \quad (3.7)$$

This formulation models the query as a unit source of semantic mass that is distributed across the selected context elements.

3.3.3 Entropic OT Formulation

The Optimal Transport cost between the query and the context set is defined as:

$$OT(q, C) = \min_{T \in \Pi(a, b)} \langle T, C \rangle + \lambda H(T) \quad (3.8)$$

where:

- T is the transport plan,
- λ is the entropy regularization parameter,
- $H(T)$ is the entropy term.

A lower Optimal Transport cost indicates better semantic alignment between the query and the selected context set.

3.4 Greedy Context Construction

The context set is constructed incrementally using a greedy strategy.

Let C_t denote the context set after t selections.

At each iteration, the next evidence sentence is selected as:

$$s^* = \arg \min_{s \in S \setminus C_t} OT(q, C_t \cup s) \quad (3.9)$$

where:

- s^* denotes the evidence sentence selected at the current step,
- S represents the set of all candidate evidence sentences,

- C_t denotes the context set at iteration t ,
- $S \setminus C_t$ represents the remaining candidate sentences,
- $OT(q, C_t \cup s)$ is the Optimal Transport cost after adding sentence s .

The selected sentence produces the maximum reduction in semantic mismatch.

3.4.1 Marginal Semantic Gain

The marginal semantic gain at iteration t is defined as:

$$\Delta_t = OT(q, C_{t-1}) - OT(q, C_t) \quad (3.10)$$

where:

- Δ_t represents the marginal semantic gain,
- q is the query embedding,
- C_{t-1} is the context set before the current selection,
- C_t is the updated context set,
- $OT(q, C)$ denotes the Optimal Transport cost.

The following properties hold:

- $OT(q, C_t)$ is non-increasing,
- $OT(q, C_t) \geq 0$.

Therefore:

$$\Delta_t \rightarrow 0 \quad (3.11)$$

which indicates that the context set gradually reaches semantic saturation.

3.5 Stopping Conditions

There are three stopping mechanisms defined :

3.5.1 Fixed-Size Constraint

Selection stops when:

$$|C| = K \quad (3.12)$$

where K is the threshold defined to stop the retrieval.

3.5.2 Adaptive Stopping

Selection terminates when:

$$\Delta_t < \epsilon \quad (3.13)$$

repeats p iterations. ϵ denotes the threshold for minimum improvement

This automatically identifies semantic saturation .

3.6 Temporal Optimal Transport

At temporal step t , the objective becomes:

$$J_t(C) = OT(q, C) + \alpha \cdot OT(C^{(t-1)}, C) \quad (3.14)$$

where:

- $OT(q, C)$ is the query relevance component ensures query coverage.
- $\alpha \cdot OT(C^{(t-1)}, C)$ is the temporal consistency component imposes temporal consistency.
- α is a trade-off parameter which balances query coverage and temporal consistency..

Temporal expansion halts when:

$$|OT(q, C^{(t)}) - OT(q, C^{(t-1)})| < \epsilon \quad (3.15)$$

This determines whether adding new segments improved the context and semantic alignment. This ensures stabilization across iterations.

3.7 Theoretical Guarantees

Theorem 3.1 (Existence of Semantic Saturation). *There exists a finite step t^* such that*

$$\Delta_t \rightarrow 0 \quad \text{for } t \geq t^*. \quad (3.16)$$

Proof. - By greedy monotonicity, the sequence $\{OT(q, C_t)\}_{t=0}^{|\mathcal{S}|}$ is non-increasing. Since $OT(q, C_t) \geq 0$, it is bounded below. Since $\mathcal{S}$ is finite, the sequence is finite and monotone, hence convergent.

Convergence of $OT(q, C_t)$ implies

$$\Delta_t = OT(q, C_{t-1}) - OT(q, C_t) \rightarrow 0. \quad (3.17)$$

□

Theorem 3.2 (Convergence Under Temporal Expansion). *Under finite candidate segmentation and greedy minimization of*

$$OT(q, C) + \alpha \cdot OT(C^{(t-1)}, C), \quad (3.18)$$

the temporal construction sequence stabilizes:

$$OT(q, C^{(t)}) \rightarrow OT(q, C^{(t-1)}). \quad (3.19)$$

Proof. - The temporal objective at step t is defined as:

$$J^{(t)} = \min_C \left(OT(q, C) + \alpha \cdot OT(C^{(t-1)}, C) \right). \quad (3.20)$$

By construction of $C^{(t)}$,

$$J^{(t)} = OT(q, C^{(t)}) + \alpha \cdot OT(C^{(t-1)}, C^{(t)}). \quad (3.21)$$

Since both transport terms are non-negative,

$$J^{(t)} \geq 0. \quad (3.22)$$

Since each step performs greedy minimization,

$$J^{(t+1)} \leq J^{(t)}. \quad (3.23)$$

Hence the sequence $\{J^{(t)}\}$ is non-increasing and bounded below by 0. Under definite candidate segmentation, only finitely many distinct context states are possible. Therefore, the sequence $\{J^{(t)}\}$ converges.

Convergence of $J^{(t)}$ implies

$$\text{OT}(q, C^{(t)}) \rightarrow \text{OT}(q, C^{(t-1)}), \quad (3.24)$$

establishing stabilization of query coverage across successive temporal states. $\square$

Property 3.1 (Diminishing Marginal Semantic Gain).

$$\mathbb{E}[\Delta_{t+1}] \leq \mathbb{E}[\Delta_t]. \quad (3.25)$$

This behavior is reminiscent of diminishing-return phenomena observed in submodular optimization.

Early selections capture significant semantic mass between the query and the candidate pool. Subsequent additions mostly overlap with already covered semantic regions, producing only smaller reductions in transport cost. This is reflected in the efficiency–effectiveness trade-off observed in **Results**, where OT-based methods achieve better coverage with compact context sizes.

Property 3.2 (Redundancy-Dominated Expansion Beyond Saturation). *Once $t > t^*$ is satisfied, further expansion increases internal similarity within C_t rather than reducing residual query mismatch.*

Let redundancy be approximated by

$$\text{Red}(C_t) = \frac{1}{|C_t|(|C_t| - 1)} \sum_{i \neq j} \cos(c_i, c_j). \quad (3.26)$$

Empirical evidence demonstrates that, beyond the point of saturation, redundancy increases as marginal gain decreases. This trend suggests reinforcement of previously addressed supports rather than the addition of complementary coverage.

Property 3.3 (Coverage Scaling with Query Complexity).

$$\mathbb{E}[|C^*(q)|] \propto |G(q)|. \quad (3.27)$$

Across datasets, adaptive OT adjusts constructed context size according to gold evidence requirements, whereas ranking-based retrieval maintains a depth k regardless of query complexity.

Property 3.4 (Single-Evidence Bias).

$$\sum_{i=2}^k \Delta_i \approx 0. \quad (3.28)$$

Dense ranking often concentrates similarity mass on a dominant sentence:

$$\cos(q, c_1) \gg \cos(q, c_i), \quad \forall i > 1.$$

Empirically, this leads to early saturation of coverage under ranking-based retrieval, particularly among multi-hop queries.

Property 3.5 (Adaptive Stopping as a Proxy for Semantic Sufficiency). *Adaptive OT halts when*

$$\Delta_t < \varepsilon \quad \text{for } p \text{ consecutive steps.} \quad (3.29)$$

This criterion applies semantic sufficiency directly in the embedding space. It stops expanding once the remaining transport mass becomes insignificant compared to the incremental gain.

3.8 Algorithm

We outline the proposed OT-guided context construction algorithm below.

Algorithm 2 Unified OT-Guided Context Construction

Require: Query q , candidate pool $\mathcal{S} = \{s_1, \dots, s_N\}$, objective function $J(\cdot)$, mode $\in \{\text{fixed}, \text{adaptive}\}$, K (if fixed), ε, p (if adaptive)

Ensure: Constructed context set C

```
1: Encode query embedding  $\mathbf{q}$ 
2: Encode candidate embeddings  $\{\mathbf{s}_1, \dots, \mathbf{s}_N\}$ 
3:  $C \leftarrow \emptyset$ 
4:  $t \leftarrow 0$ 
5: patience  $\leftarrow 0$ 
6: while  $\mathcal{S} \setminus C \neq \emptyset$  do
7:    $t \leftarrow t + 1$ 
8:   for all  $s \in \mathcal{S} \setminus C$  do
9:     Compute  $J(s) = J(C \cup \{s\})$ 
10:   end for
11:    $s^* \leftarrow \arg \min_{s \in \mathcal{S} \setminus C} J(s)$ 
12:   prev_cost  $\leftarrow J(C)$ 
13:    $C \leftarrow C \cup \{s^*\}$ 
14:   new_cost  $\leftarrow J(C)$ 
15:    $\Delta_t \leftarrow \text{prev\_cost} - \text{new\_cost}$ 
16:   if mode = fixed and  $|C| = K$  then
17:     break
18:   end if
19:   if mode = adaptive then
20:     if  $\Delta_t < \varepsilon$  then
21:       patience  $\leftarrow \text{patience} + 1$ 
22:     else
23:       patience  $\leftarrow 0$ 
24:     end if
25:     if patience  $\geq p$  then
26:       break
27:     end if
28:   end if
29: end while
30: return  $C$ 
```

Algorithm 3 Temporal OT Context Construction

Require: Query q , segmented pool $\{\mathcal{S}^{(1)}, \dots, \mathcal{S}^{(T)}\}$, temporal weight α, ε, p

Ensure: Context states $\{C^{(1)}, \dots, C^{(T^*)}\}$

```
1:  $C^{(0)} \leftarrow \emptyset$ 
2: for  $t = 1$  to  $T$  do
3:   Define temporal objective
      
$$J_t(C) = \text{OT}(q, C) + \alpha \cdot \text{OT}(C^{(t-1)}, C)$$

4:   Apply Algorithm 2 on  $\mathcal{S}^{(t)}$ 
5:   using objective  $J_t(\cdot)$  in adaptive mode with parameters  $\varepsilon, p$ 
6:   Obtain  $C^{(t)}$ 
7:   if  $|\text{OT}(q, C^{(t)}) - \text{OT}(q, C^{(t-1)})| < \varepsilon$  then
8:     break
9:   end if
10: end for
11: return  $\{C^{(1)}, \dots, C^{(t)}\}$ 
```

3.9 Summary

This chapter presented the mathematical formulation of the proposed OT-guided context construction framework. The framework is training-free, theoretically grounded, and suitable for reasoning-oriented retrieval tasks.

System Design

4.1 System Design Overview

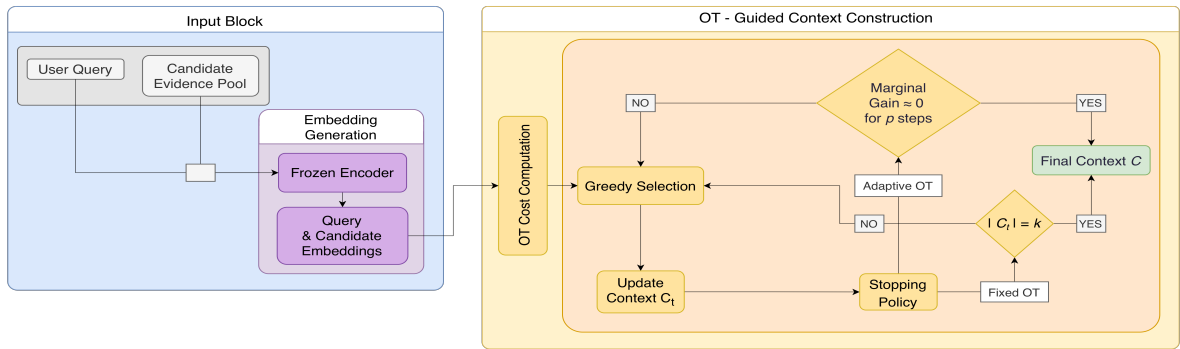


Figure 4.1: OT-Guided Context Construction Architecture.

Figure 4.1 represents the proposed Optimal Transport (OT)-based architecture for context construction. The system begins with an Input Block that handles the query along with a pool of potential candidates. Followed by a frozen encoder that creates dense semantic embeddings of both query and candidate supports. These embeddings are then sent to the OT Cost Computation module which compares the semantic transport cost between query distribution and the candidate evidence.

The OT-Guided Context Construction module is the central part of the architecture. The module picks candidate sets through a greedy selection process based on the Optimal Transport objective. Following each selection, the changing context is adjusted and screened using a Temporal OT Objective that ensures semantic correspondence and consistency throughout the time-steps. A stopping policy in place decides whether the selection process should be stopped or not.

The system can be used with either a Fixed OT strategy, which terminates once a specified context size has been achieved, or with Adaptive OT strategy, which terminates selection once the marginal improvement in the OT objective lies within customarily insignificant values. When the stopping condition is met, the system gives out the best optimized context.

4.1.1 Input Module

The input module converts user query and candidate evidence into structured semantic forms . The process is initiated by creating a single dense embedding of the query , the candidate evidence pool provides support segments that can be added to the final context. The query and candidate segments are then processed by a frozen encoder which, provides a consistent and stable embedding . The encoder generates sparse-dense representations of the query and the entire candidate segments, which is used for future similarity and cost calculations .

4.1.2 Embedding Generation Module

The Embedding Generation Module encodes all the input in a single semantic space. Through the use of a frozen encoder, the system is able to inhibit the drift in the parameters during iterations , thus remaining stable .The embeddings generated are sent to the OT Cost Computation module.

4.1.3 OT Cost Computation Module

This module is used to compute the transport cost between the query embedding and the candidate evidence embeddings. The system solves a model of the alignment between the query distribution and the distribution of candidate segments. This is done to measure the effectiveness with which the candidate information covers the query. The cost matrix resulting will indicate the contribution made by each candidate segment in satisfying the information requirement in the query. These calculated costs are direct guiding factors in the greedy selection stage that follows .

4.1.4 OT-Guided Construction Module

This is an iterative process that selects the candidate segment that minimizes the existing OT objective. Upon selection, the segment is added to the current state of evolving context which is denoted as C_t . The Temporal OT Objective measures the effectiveness of the expanded context to enhance the correspondence to the query without affecting the consistency between the previously chosen segments.

A stopping policy controls the termination. With the Fixed OT strategy, the process stops when the context is at a predetermined size constraint. The Adaptive OT strategy, in contrast, observes the marginal gain in the OT goal over successive steps and halts the process when further segments are not leading to substantial gain. This process of refinement guarantees that only the semantically useful, non redundant pieces of information are incorporated in the final constructed context.

4.1.5 Temporal Optimal Transport Mechanism

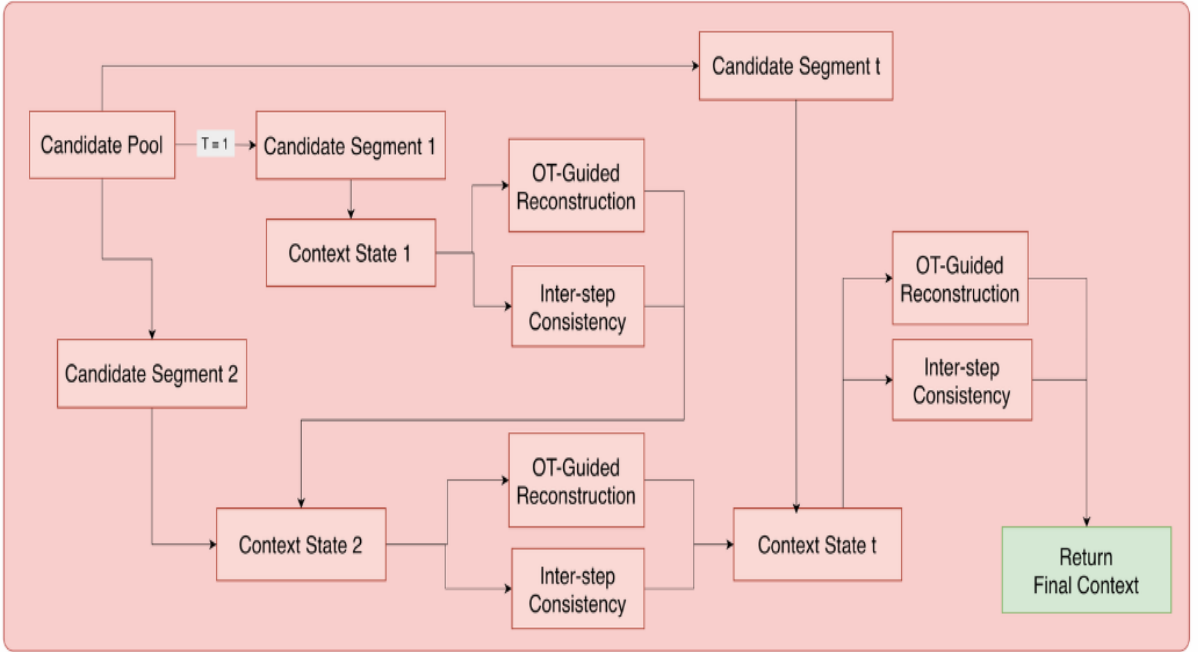


Figure 4.2: Temporal OT.

Figure 4.2 illustrates the context evolution process by Temporal OT.

The process starts with the candidate pool from which a segment is chosen and constitutes the first context state. The OT-Guided Reconstruction mechanism measures the effectiveness with which this context is able to satisfy the query.

Iteratively, a new candidate segment is chosen and added to the context under evolution leading to new context states. In this process, inter-step consistency is imposed so that it avoids repetition and semantic drifting. The Temporal OT Objective ensures cumulative transport improvement, so that every addition to it significantly improves query alignment.

The process repeats until coverage saturation is achieved. The saturation could be detected when the marginal OT gain is small or when inter-step semantic similarity shows that there is redundancy between the chosen segments. In such conditions, the system gives back the final context. Otherwise, the process continues with the evaluation of other candidates. This guarantees progressive semantic coverage, less redundancy and principled termination.

4.1.6 Stopping and Final Context Generation

The system will produce the final context C , on fulfilling any of the below stopping conditions : fixed cardinality constraint, marginal OT gain threshold or coverage saturation detection.

4.1.7 Key Design Advantages

Through the combination of Optimal Transport cost computation and temporal consistency modeling, the system will be capable of semantically accurate and redundancy conscious selection of candidate evidence.

4.2 Design Diagrams

4.2.1 Data Flow Diagram(DFD)

The Data Flow Diagram shows how data is transported on a high-level basis between the main modules of the OT-directed context construction system. The process starts with the User making a query and, as illustrated in Figure 4.3, ends with the system responding to the query. The query is sent to the Query Encoding subsystem, where it is

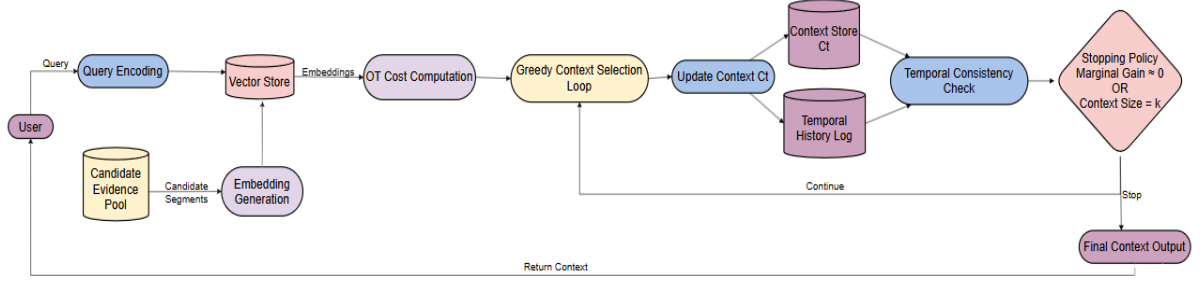


Figure 4.3: Data Flow Diagram

converted into a dense semantic embedding. At the same time, the Candidate Evidence Pool provides candidate segments which are processed by Embedding Generation so that both the query and candidates are projected into a common vector space.

These embeddings are stored in the Vector Store and forwarded to the OT Cost Computation module where the transport cost is computed. The resulting cost matrix drives the Greedy Context Selection Loop, which repeatedly selects the most semantically compatible candidate segment.

Once a segment is selected, the Update Context Ct module incorporates it into the evolving context state. The Stopping Policy module determines whether to continue the iterative selection process. If the stopping criteria are not satisfied, the system repeats the greedy selection process; otherwise, the Final Context Output is generated and returned to the User. This structured data movement ensures cost-aware, temporally stable, and redundancy-controlled context construction..

4.2.2 Sequence Diagram

The Sequence Diagram represents the dynamic correlation between components of the system. The process starts when the User enters a query in the Input Block and is as illustrated in Figure 4.4. Followed by Frozen Encoder that produces semantic embeddings of all inputs and sends them back to the OT Engine which triggers OT Cost Computation and gets into an iterative loop that would run till the stopping policy condition is met.

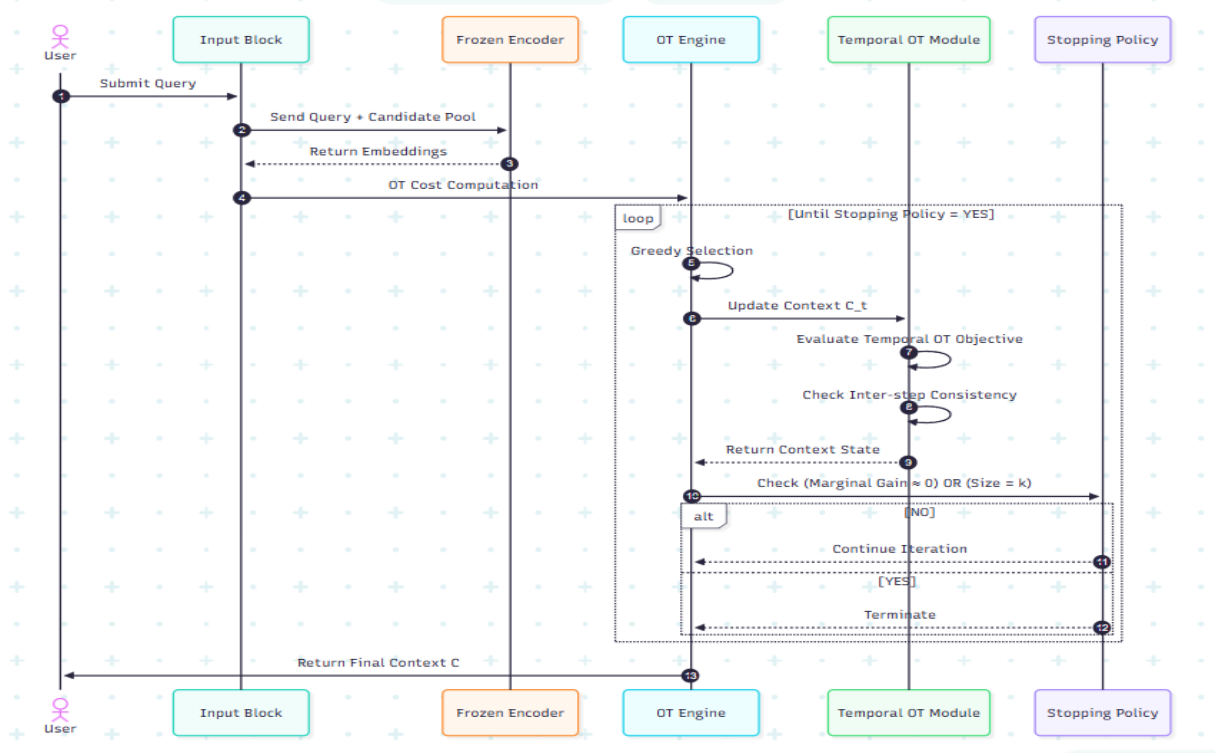


Figure 4.4: Sequence Diagram

Each iteration is followed by Greedy Selection of the candidate segment that optimizes the present transport objective . Temporal OT Module helps avoid any redundancy or semantic drift when new segments are added. The revised context state is in turn sent back to the OT Engine .When the stopping condition turns out to be false then the system repeats the iteration loop. When its true, the process is concluded and the final optimal context C is sent back to the User.

4.3 Algorithms Used

This section details the core algorithms and concepts used in the system.

4.3.1 Sinkhorn Algorithm [6]

Core Idea:

The Sinkhorn algorithm is an effective approach to the entropically regularized Optimal Transport (OT) problem. The alternative to resolving the classical OT formulation with linear programming is the addition of an entropy regularization term to the transport objective. This makes the optimization convex and computationally effective.

Iterative matrix scaling, also referred to as Sinkhorn iterations is used to find the solution.

Explanation:

The entropy regularization solves the optimal transport problem by smoothing and by allowing a rapid convergence by updating the row and column normalization. The Sinkhorn iterations are much more computationally complex and can scale to large embedding spaces in comparison to classical OT solvers. The Sinkhorn algorithm is employed in this project to calculate the OT distance between query embeddings and candidate context embeddings, which serves as a structured similarity measure.

Application in the Project:

The OT distance calculated is a metric of the similarity between the distribution of candidate context embeddings and query embedding distribution. This transport cost is the one that is minimized in the context building.

Steps:

- Build the cost between the query embedding and candidate embeddings in a cost matrix.
- Add entropy to regularize the transport problem.
- Make sinkhorn updates that is , repeated row and column normalizations.
- Repeat until marginal constraints are converged.
- Calculate the final OT distance of the resulting transport plan.

4.3.2 Greedy Forward Selection [7]

Core Idea:

Greedy forward selection is an iterative subset construction algorithm that builds a solution by successively adding into it the element that optimizes the most marginal improvement of the objective function. Recent theoretical findings indicate that greedy algorithms can give good approximation guarantees provided that the objective function is submodular.

Explanation:

Each iteration of the algorithm assesses the marginal value of each candidate element being added to the existing set and the one providing the largest decrease in the objective is chosen. This gradual process of construction guarantees a diminishing marginal gains as well as preservation of computational efficiency. In addition to strong submodularity, in practice empirical diminishing returns can frequently support greedy selection.

Application in the Project:

The context set is built up by repeatedly choosing the candidate passage which minimizes the Optimal Transport objective on addition to the existing context set. Officially, in every step, the algorithm picks the candidate yielding the minimum OT distance between query and the updated context set. This will lead to gradual enhancement in relevance contexts.

Steps:

- Initialise context set empty.
- Calculate the OT objective of each candidate pass upon its addition to the existing set.
- Choose the candidate with the greatest minimization of OT distance.
- Include the successful candidate into the context.
- Repeat till termination criterion is attained.

4.3.3 Adaptive Threshold Stopping [8]**Core Idea:**

Adaptive threshold stopping puts a stopping condition into greedy optimization. The process of the selection is terminated when the marginal improvement of the objective becomes less than a specified threshold. This guarantees the efficiency in computations and avoids redundant additions.

Explanation:

The threshold-based greedy frameworks adaptively regulate the construction of the subsets by comparing the marginal gains to a threshold value. Once the gains become insignificant the process will stop. This finds a compromise between solution quality and scalability particularly in large candidate spaces.

Application in the Project:

The greedy algorithm is stopped once the marginal decrease in Optimal Transport distance is less than some specified threshold. This eliminates excess expansion of the context set and provides effective context-building.

Steps:

- Calculate the marginal change of distance between OTs and the addition of a candidate.
- Compare the incurred marginal gain with the predetermined level.
- In case the gain is less than the threshold, then interrupt the selection.
- Or else continue the greedy selection.

4.3.4 Temporal regularization [9]**Core Idea:**

Temporal regularization is a form of regularization that builds on top of other regularization methods (such as the static optimization) in order to introduce stability between sequential iterations. Rather than optimizing the individual steps, a penalty term is introduced in order to promote consistency of the current solution with the previous solution.

Explanation:

Sequential optimization provide stable constraints that forbids sudden shifts between runs. The solution is smoothly optimized by introducing a regularization term , by not awarding rewards to the previous state, yet solves the primary objective. It is mostly applicable in dynamic systems or iterative systems.

Application in the Project:

The objective function involves the use of a time stability term which balances relevance of queries with stability over iterations. The optimization objective is a combination of the OT distance between the query and the current context with a penalty term of the distance between the previous context and the current context. One of the hyperparameters is what determines the trade-off between relevance and stability.

Steps:

- Store context from the last iteration.
- Formulate an augmented objective, containing a time-dependent penalty.
- Greedy selection under modified objective.
- The next iteration of context can be updated.

4.4 Requirements

4.4.1 Software Requirements

Software/Frameworks

- Operating System: Linux / Windows
- Programming Language: Python 3.9+

Libraries & Tools:

- PyTorch – embedding computation
- HuggingFace Transformers – frozen encoder (BGE-base-en-v1.5)
- FAISS – dense candidate shortlisting
- NumPy / SciPy – numerical computation
- POT / custom Sinkhorn implementation – Optimal Transport
- Scikit-learn – evaluation utilities

4.4.2 Hardware Requirements

- GPU (NVIDIA CUDA-enabled)
- Minimum 16 GB RAM
- Multi-core CPU
- Storage (50–100 GB recommended)

4.5 Project Timeline and Work Division

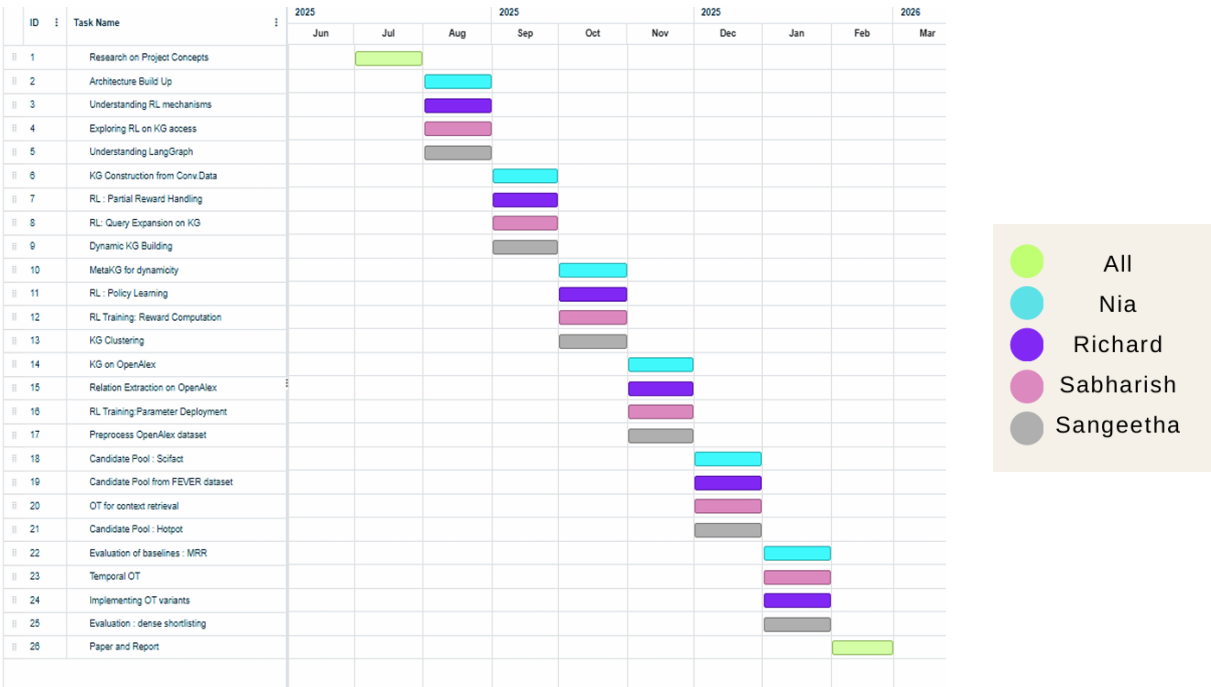


Figure 4.5: Gantt Chart

4.6 Summary of Chapter

This chapter provides a comprehensive blueprint for system implementation, ensuring clarity in design, functionality, and resource planning.

Chapter 5

Results and Discussions

5.1 Experimental Setup

The Optimal Transport (OT) based context construction framework was tested on three benchmark datasets: HotpotQA (multi-hop question answering)[10], SciFact (multi-evidence scientific claim verification)[12], and FEVER (fact verification)[11]. These sets were chosen to assess performance on reasoning intensive tasks.

Three variants of OT baselines, namely: OT-Fixed (fixed context size), OT-Adaptive (dynamic stopping relying on marginal semantic gain) and OT-Temporal (learns in multiple steps) were analyzed.

The following measures were considered :

- Hit@ Set : This is used to measure how many gold evidence sentences are contained within the set of sentences retrieved.
- Average Gold Evidence Retrieved (Avg Gold) : is the average number of sentences of the supporting evidence retrieved.
- Average Context Size : denotes the number of sentences in the constructed final context.

A combination of all these metrics permits measuring effectiveness (retrieval quality) as well as efficiency (compactness of context).

5.2 Effectiveness Analysis

Table 5.1: Effectiveness Comparison Across Datasets

Method	HotpotQA		SciFact		FEVER	
	Hit@Set	Avg Gold	Hit@Set	Avg Gold	Hit@Set	Avg Gold
Dense	0.10	1.66	0.98	1.12	1.00	2.92
OT-Fixed	0.95	2.00	0.99	1.10	1.00	1.30
OT-Adaptive	0.86	2.00	0.96	2.00	0.99	2.00

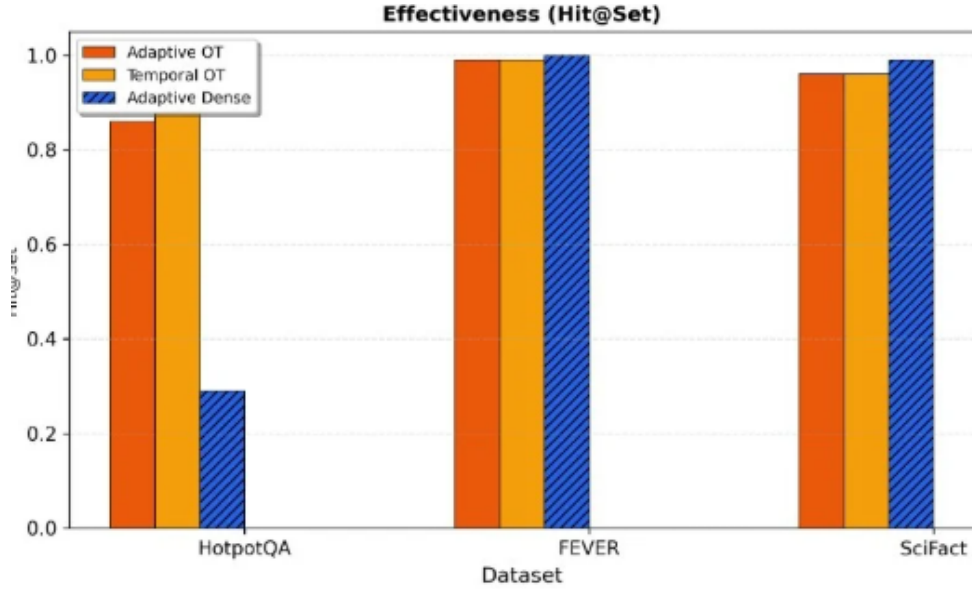


Figure 5.1: Effectiveness (Hit@Set) comparison across datasets.

An objective of the analysis is to determine whether evidence recovery is improved in a semantic coverage optimization model compared to ranking based Dense retrieval model.

Dense retrieval on HotpotQA, a multi hop reasoning task, has a Hit@Set of 0.10, so it does not recover extensive supporting information. However, the performance is greatly enhanced to 0.95 by OT-Fixed. OT-Adaptive is also strong, with a dynamic size adjustment and high effectiveness, which is 0.86. This shows significant enhancement in multi-hop evidence recovery .

On SciFact, there is consistent performance across all the methods . OT-Adaptive enhances the average gold evidence sentences retrieved from 1.12 to 2.00 . This implies that OT-based selection retrieves complementary evidence and not redundant sentences.

In FEVER, where most claims take a single supporting sentence , performance is stable at around 1.00 Hit@Set . This shows that OT-based retrieval itself does not impose any cost on simpler tasks and also shows significant benefits on more reasoning-intensive datasets.

Overall, its observed from Table 5.1 and and Figure 5.1 that the set-level semantic coverage problem interpretation of retrieval can greatly enhance performance in a multi-hop and multi-evidence problem.

5.3 Efficiency Analysis

Table 5.2: Efficiency Comparison on HotpotQA

Method	Hit@Set	Avg. Context Size
Adaptive Dense	0.29	33.89
OT-Adaptive	0.86	3.39

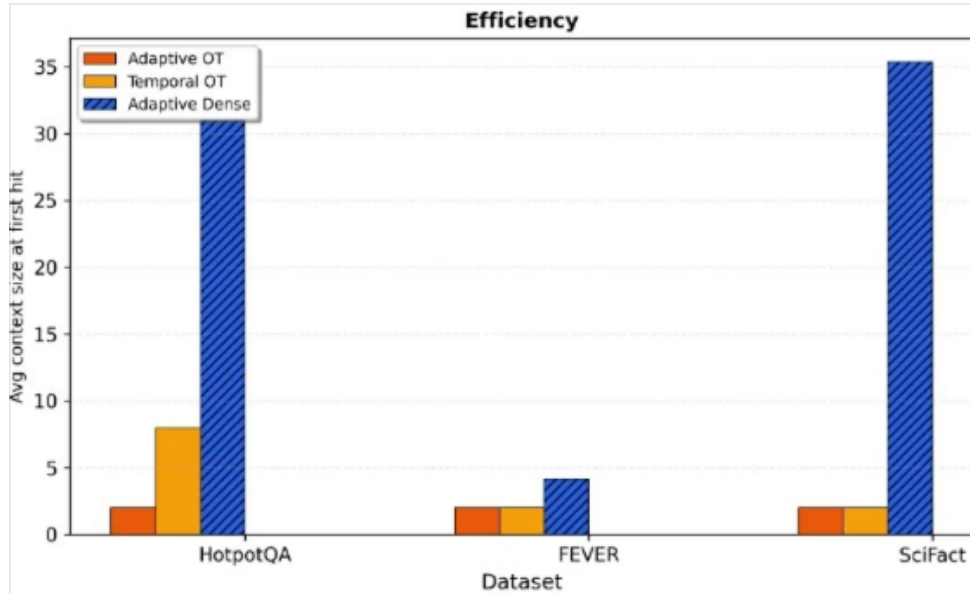


Figure 5.2: Efficiency in terms of average context size .

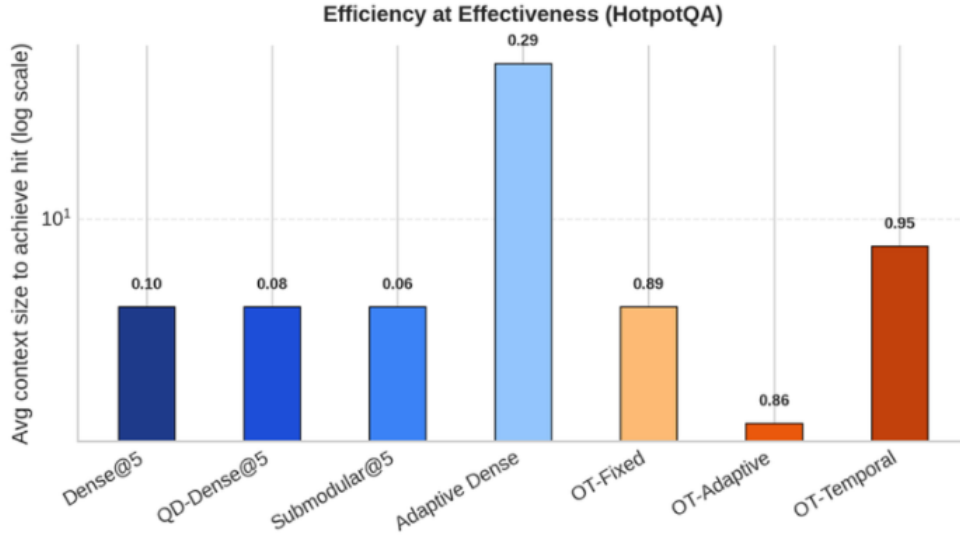


Figure 5.3: Efficiency at Effectiveness (HotpotQA).

Besides being effective, the retrieval systems should also be efficient. Expanding the amount of recalled sentences in Dense retrieval tends to create redundancy as opposed to higher semantic adequacy.

Dense retrieval on HotpotQA takes 33.89 sentences to get a small Hit@Set of 0.29. Conversely, OT-Adaptive has a Hit at Set of 0.86 and it uses a mere 3.39 sentences. This is almost a 90 % reduction in the context size and at the same time, enhances effectiveness.

Likewise on the SciFact, OT-Adaptive minimizes the context size needed by about 94 percent of the methods of Dense retrieval.

The findings in Table 5.2, Figure 5.2, and Figure 5.3 indicate that OT-based retrieval is more effective and has a lower context size. This means that it is more efficient-effective than the conventional retrieval which is based on ranking.

5.4 Coverage Beyond Single Evidence

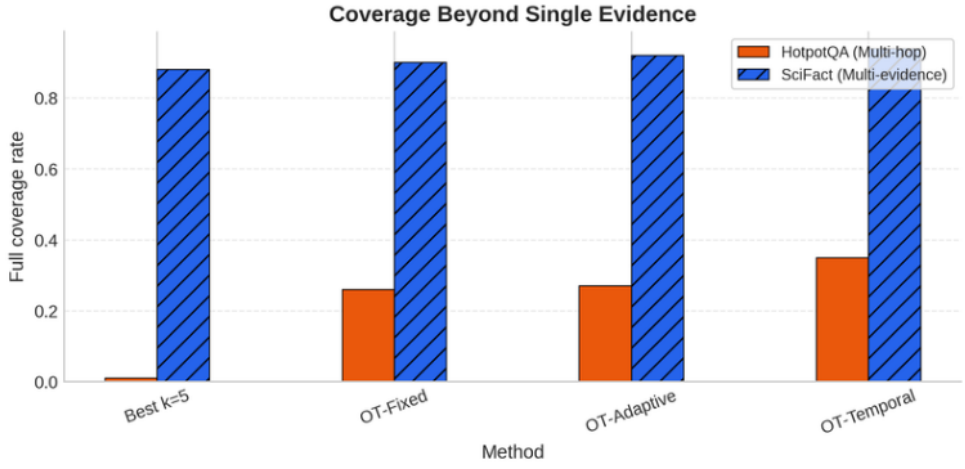


Figure 5.4: Coverage beyond single evidence.

Complete coverage of evidence was considered to further analyse the quality of reasoning.

Dense@5 is only able to cover 0-11% of the full evidence in multi-hop reasoning in HotpotQA. OT-Fixed and OT-Adaptive are fully covered to 26-27%. OT-Temporal enhances the coverage to 35%, which is an almost three times better than Dense retrieval. In the case of SciFact, the coverage increases to 0.95 .

These results in Figure 5.4 indicate that OT-based retrieval is not only more accurate, but can also be more semantically sufficient.

5.5 Discussion

5.5.1 OT enhances Multi-Hop Retrieval.

The results on HotpotQA demonstrate an essential weakness of ranking-based retrieval systems. Dense retrieval compiles sentences according to similarity scores. Multi-hop reasoning needs complementary evidence to answer a query.

The OT framework minimizes the residual semantic mismatch between the query and constructed context set. Sentences are selected at every step according to their marginal contribution to lowering the overall transport cost which guarantees that only new semantic information is added. The better performance on multi-hop tasks is due to the shift from independent ranking to set-level optimization.

5.5.2 Stability on Single hop tasks

On FEVER, where single-hop claims are the main contributor, OT-based approaches achieve the same performance as Dense retrieval. This shows that semantic coverage modeling can help in boosting complex reasoning activities without affecting simple retrieval situations.

Therefore, OT based context construction is robust to various levels of tasks.

5.5.3 Adaptive Stopping efficiency gains

The standard retrieval methods tend to enhance recall , thereby creating larger and overlapping context pools.

Meanwhile , a diminishing marginal semantic gain stopping mechanism is proposed in OT-Adaptive. Retrieving ceases automatically when no more sentences can be included in the retrieval that cause a significant decrease in the transport cost.

The outcome will be a reduced but semantically complete context set .

5.5.4 Impact of Temporal OT

Temporal OT imposes inter-step semantic consistency across context states , imposing stability and coherence across retrieval steps rather than making evidence selection a one-step optimization.

This leads to increased full-evidence recovery on multi-hop tasks as can be shown by 35% coverage of HotpotQA.

5.6 Summary of Findings

The results confirm the main hypothesis of this work : reasoning tasks show a better quality of retrieval when they are modeled as a set-level semantic coverage maximization problem, and not as a ranking problem.

The proposed model enhances multi-hop evidence recovery, improves multi-evidence completeness, reduces redundancy, achieves higher efficiency gains. and also shows stability in single-hop tasks. These results show that Optimal Transport gives a principled and efficient method of context construction for reasoning intensive information retrieval systems.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

The project addresses a limitation of conventional ranking based information retrieval systems, which is that they lack in retrieving complementary evidence needed to complete reasoning-based queries. To address this shortcoming, the proposed framework redefines retrieval as a problem of context construction and evaluates semantic coverage between queries and candidate evidence by applying Optimal Transport. The system builds context by selecting candidate sentences that minimize the cost of transportation between the query and the context set.

Experimental analysis demonstrates that the OT-based approach can be evaluated as more effective in terms of semantic coverage and has a smaller and more efficient context set than traditional dense retrieval procedures. Adaptive stopping ensures efficiency by changing the size of context depending on the complexity of the query, whereas temporal extension helps to process progressive reasoning in more than one evidence step.

6.2 Key Contributions

This project introduces a paradigm to enhance the context-building of reasoning-based information retrieval systems. This key contribution of this project lies in reformulating retrieval as a context construction problem, where the aim is to assemble a concise set of evidence that jointly fulfills the semantic requirements of a query, rather than retrieving documents based on ranking. It proposes a semantic coverage framework using OT that quantifies the semantic mismatch between a query and the constructed context set. A greedy OT-guided method of constructing context is suggested to successively choose candidate sentences depending on their marginal contribution to the semantic coverage and redundancy reduction.

The framework also presents three types of context construction, namely, Fixed-Size OT, Adaptive OT through coverage-based stopping, and Temporal OT in progressive evidence aggregation when performing multi-hop reasoning problems. The variants enable the system to support varying constraints of retrieval and query complexity.

6.3 Impact of the Proposed System

The proposed framework enhances the retrieval systems employed in the reasoning based applications by ensuring that all the evidence which is retrieved meets the semantic demands of a query. The system reduces redundancy and increases the quality of the context retrieved by concentrating on set-level semantic coverage.

This method proves especially useful with LLMs and RAG systems, in which the size of the context is constrained by token usage.

6.4 Limitations

The computation of OT mechanism has an additional overhead over existing similarity-based retrieval strategies . Another limitation is the use of pretrained embeddings of a frozen encoder. Given that the encoder is not domain-specific, embeddings might not necessarily induce domain-specific semantic relations.

6.5 Future Scope

Future research can focus on improving the scalability of the framework by coming up with effective approximations of OT calculations. It can also be enhanced by combining the OT-based context construction mechanism with LLM reasoning pipelines, including RAG systems, to make the generated responses more reliable. By integrating the framework with structured knowledge sources like KGs, it can be possible to improve reasoning on complex multi-hop queries. The system can potentially learn to adapt the semantic coverage indicators of particular domains and applications based on learning-based extensions.

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Appendix A: Presentation

Context Construction for Reliable Reasoning in High-Stakes Information Retrieval using Optimal Transport

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Project Guide:

Ms. Mehbooba P Shareef

Contents

01	Problem Definition	09	Assumptions
02	Project Objective	10	Work Breakdown
03	Purpose and Need	11	Requirements
04	Literature Survey	12	Gantt Chart
05	Proposed Method	13	Risks and Challenges
06	Architecture Diagram	14	Conclusion
07	Design Diagrams	15	References
08	Results & Output		

Introduction

Modern AI systems rely heavily on effective context retrieval and ranking. Existing retrieval pipelines struggle with contextual relevance in complex queries. Improving contextual quality is critical for downstream reasoning and decision systems.

This project explores an *Optimal-Transport (OT)* based retrieval approach for improved context quality.

Instead of answering whether **text A** and **text B** are similar, we find how hard it is to reshape **A** to **B** by leveraging the fact that meaning is distributed and does not rest at one point.

Problem Statement

To build a retrieval mechanism that improves contextual relevance without task-specific training by answering the question: *Can the meaning of **Q** be explained efficiently using the meaning resting within **C**?*

Current retrieval systems depend largely on dense or sparse similarity scores

Fails to capture fine-grained contexts from documents

Top-ranked results are often redundant

Primary Goal

To develop a *training-free, OT* based *context construction* system that retrieves a ***minimal, semantically sufficient set of evidence*** for reasoning-oriented information retrieval tasks.

Key Objectives

- Reformulate context retrieval as set construction problem
- Implement OT based semantic coverage modelling
- Design adaptive stopping mechanism
- Develop temporal OT extension for multi-hop reasoning
- Evaluate effectiveness and efficiency across benchmark datasets

Purpose

To build a retrieval system that ensures semantic coverage, not just similarity ranking. The system constructs a minimal, non-redundant context set that fully explains the query.

Need

- Ranking-based retrieval does not ensure complete evidence coverage.
- Increasing Top-K adds redundancy, not semantic sufficiency.
- Multi-hop and reasoning tasks require complementary evidence.
- No existing training-free method is optimised for set-level semantic coverage.

Literature Survey

Dense Passage Retrieval for Open-Domain Question Answering (DPR, 2020)

Citation: Karpukhin et al., 2020

Core Idea

- Introduced dual-encoder dense retrieval for open-domain QA.
- Represents questions and passages as dense semantic vectors.

Contribution

- Moves beyond keyword matching (BM25).
- Significant improvement in semantic retrieval accuracy.
- Established dense retrieval as a strong baseline.

Limitations

- Operates as independent ranking.
- Does not explicitly model set-level coverage.
- May retrieve redundant but individually similar passages.

Relevance-Aware Contrastive Pre-Training (RACP, 2023)

Citation: Lei et al., 2023

Core Idea

- Unsupervised dense retrieval training without labeled relevance data.
- Uses relevance-aware contrastive learning.

Contribution

- Reduces dependency on expensive annotations.
- Improves embedding quality using pseudo positive/negative pairs.
- Scales to large unlabeled corpora.

Limitations

- Still ranking-based retrieval.
- Does not ensure semantic sufficiency of the final context set.
- No mechanism for adaptive context construction.

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (RAG, 2020)

Citation: Lewis et al., 2020

Core Idea

- Combines neural retriever with generative model.
- Conditions answer generation on retrieved passages.

Contribution

- Integrates parametric and non-parametric memory.
- Enables knowledge-intensive question answering.
- Improves factual grounding in generation.

Limitations

- Retrieval remains top-k ranking.
- Context size is fixed.
- No explicit modeling of evidence coverage.

Retrieve What You Need: A Mutual Learning Framework for Open-Domain Question Answering

Citation: Wang et al., 2024

Core Idea

- Introduces a knowledge selector between retriever and reader.
- Uses reinforcement learning to select relevant passages.

Contribution

- Reduces redundant passages.
- Improves retrieval and QA jointly.
- Does not require additional annotated passage labels.

Limitations

- Requires complex training and optimization.
- Selection depends on reader reward.
- Does not explicitly optimize semantic coverage.

Word Mover's Distance (WMD, 2015)

Citation: Kusner et al., 2015

Core Idea

- Uses OT to measure semantic distance between documents.
- Computes minimum cost to align word embeddings.

Contribution

- Captures semantic similarity beyond exact word overlap.
- Strong theoretical foundation in OT.

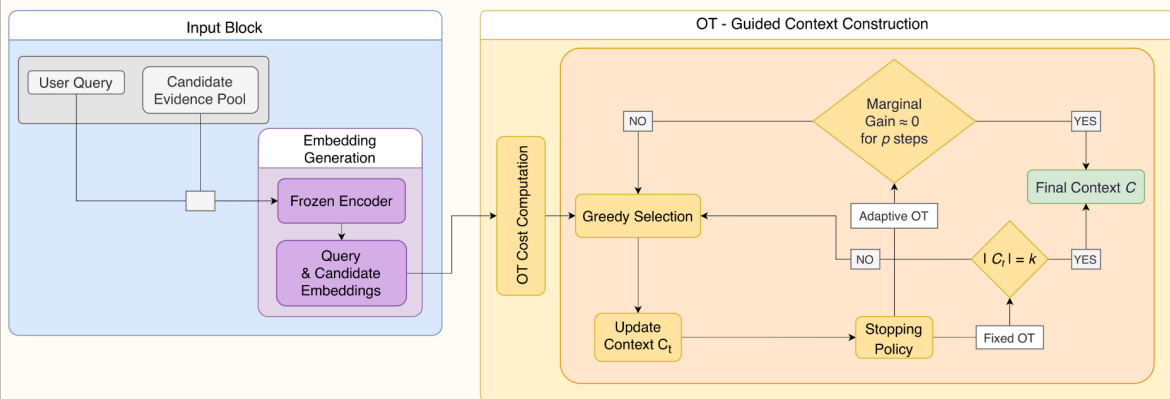
Limitations

- Designed for similarity measurement, not retrieval pipelines.
- Computationally expensive for large-scale search.
- Does not model adaptive context construction.

Paper	Method Type	Coverage Modeling	Redundancy Control	Training Required	Limitation
DPR [6]	Dense Retrieval	✗	✗	✓	Redundant ranking
RACP [3]	Dense Retrieval	✗	✗	✓	No context construction
RAG [10]	Retriever-Generator	✗	✗	✓	Fixed top-k
Mutual Learning [7]	RL Retrieval	Partial	Partial	✓	Complex training
WMD [8]	OT Similarity	✗	✗	✗	Not scalable
Proposed OT	OT Context Construction	✓	✓	✗	Higher compute

Architecture Diagram

Fixed/Adaptive OT



Fixed/ Adaptive OT

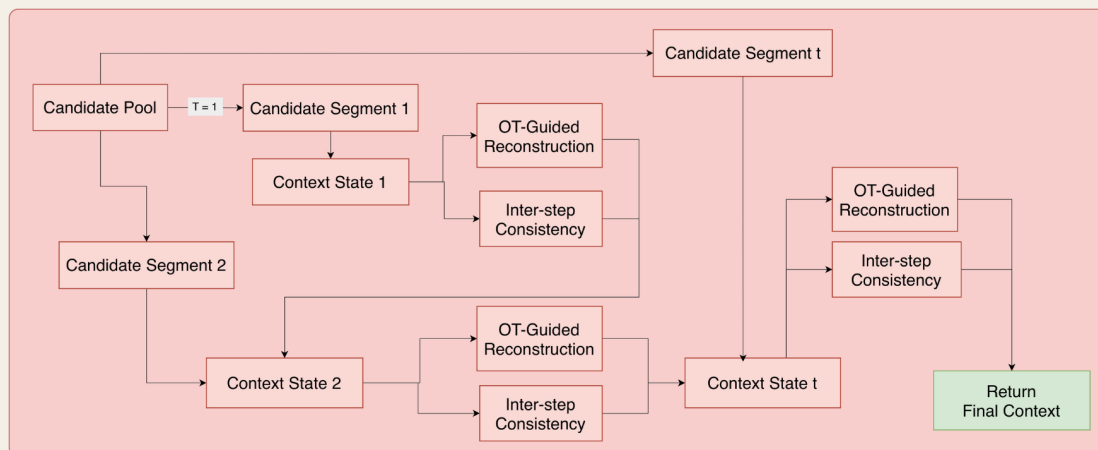
Input Block

- User *query* and *candidate evidences* are the inputs
- Query encoded as a single semantic embedding or optionally decomposed into semantic supports
- Candidate evidences are encoded using the same frozen encoder
- All embeddings lie in a shared semantic space

OT-Guided Context Construction

- Measure semantic coverage using OT
- Greedily select evidence that reduces transport cost
- Update context set iteratively
- Stop when coverage saturates or fixed k limit is attained
- Output a semantically sufficient context set

Temporal OT



Temporal OT

- Start with global candidate evidence pool
- Process candidates in ordered segments over time
- Build an initial context from the first candidate segment
- Use OT-guided reconstruction to improve semantic coverage at each step
- Enforce inter-step semantic consistency between consecutive context states
- Iteratively update the context as new candidate segments are considered
- Check for coverage saturation and semantic similarity across steps
- Return the final context set

Proposed Method

01. Problem Reformulation

Retrieval as Set Coverage

Traditional approach ranks sentences individually using cosine similarity

$$\text{score}(s) = \cos(q, s)$$

Fails for multi-fact retrieval, a single sentence rarely covers all query aspects

Our approach focuses on constructing a context set that jointly explains the query

Goal: Find a context set $C \subseteq S$ that minimizes $OT(q, C)$

achieving dual optimisation goals, i.e.,

- semantic sufficiency
- minimal redundancy

02. Representation

Representation and Cost Geometry

Embedding space

Query and sentences: $q_i, s_j \in \mathbb{R}^d$

Encoder Model: **BAAI/bge-base-en-v1.5**

Normalisation: $\|q_i\| = \|s_j\| = 1$

L2 normalization ensures similarity depends primarily on vector direction rather than magnitude.

Semantic Distance (Cost): $c_{i,j} = 1 - \cos(q_i, s_j)$

Lower cost indicates better semantic match between query aspect and sentence

03. Example

Example Query: Where was the stone used to build Threave Castle sourced and which river surrounds it?

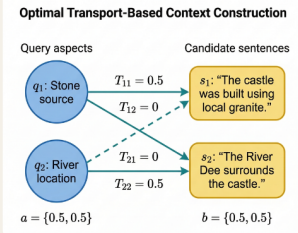
Let q_1 - stone used to build Threave Castle sourced
 q_2 - which river surrounds it

and

Candidates be s_1 - The castle was built using local granite.
 s_2 - The River Dee surrounds the castle.

Cost Matrix:

$C_{i,j}$	s_1	s_2
q_1	0.10	0.60
q_2	0.70	0.12



04. Optimal Transport

Optimal Transport as Coverage Measure

Mass Transport Framework

Query Distribution: $a = [0.5, 0.5]$

Candidate Capacity: $b = [0.5, 0.5]$

Transport Plan:

$T_{i,j}$	s_1	s_2	$\sum_i T_{ij} = a_i$
q_1	T_{11}	T_{12}	0.5
q_2	T_{21}	T_{22}	0.5
$\sum_i T_{ij} = b_j$	0.5	0.5	

04. Optimal Transport

Optimal Transport as Coverage Measure

Optimal Transport Cost: $OT(q, C) = \sum_{i,j} T_{ij} \cdot C_{ij}$

Objective: *minimize* $T_{11} \cdot 0.10 + T_{12} \cdot 0.60 + T_{21} \cdot 0.70 + T_{22} \cdot 0.12$

Transport
Plan:

$T_{i,j}$	s_1	s_2	$\sum_i T_{ij} = a_i$
q_1	0.5	0	0.5
q_2	0	0.5	0.5
$\sum_i T_{ij} = b_j$	0.5	0.5	

OT Cost:

$$OT(q, C) = 0.5(0.10) + 0.5(0.12) = 0.11$$

05. Greedy Method

Greedy OT-Guided Context Construction

Optimal subset search requires evaluating all possible sentence combinations

Each sentence can either be included or excluded from context set

Therefore, 2^N combinations (for N sentences)

Iterative Greedy Approach: $s^* = \arg \min_{s \in S} OT(q, C \cup \{s\})$

At each step, the sentence that most improve semantic coverage is selected

Marginal Semantic Gain: $\Delta t = OT(q, C_{t-1}) - OT(q, C_t)$

measures how much the new sentence improves coverage. Higher Δt indicates more valuable information.

$\Delta t < \epsilon$, for p consecutive steps => **semantic saturation**

Return final context set

Results & Output

Results

Fixed-budget performance at $k = 5$ across datasets.

Method	HotpotQA			FEVER		SciFact		
	Hit@5	Avg Gold	MH-Full	Hit@5	Avg Gold	Hit@5	Avg Gold	ME-Full
Dense	0.10	0.10	0.10	1.00	2.92	0.98	1.12	0.99
Dedup Dense	0.09	0.09	0.10	1.00	1.25	1.00	1.00	0.88
QD-Dense	0.08	0.09	0.11	0.95	1.22	0.96	0.96	0.84
BM25	0.07	0.08	0.10	0.96	2.08	0.84	0.94	0.83
MMR	0.03	0.03	0.10	0.97	1.58	0.96	0.96	0.84
Submodular	0.06	0.06	0.10	0.92	0.99	0.85	0.85	0.73
Cross-Encoder	0.07	0.07	0.10	1.00	1.17	0.92	1.05	0.92
OT-Fixed (decomp)	0.89	1.41	0.34	0.90	2.01	0.98	2.08	1.00
OT-Fixed (no decomp)	0.95	1.66	0.48	1.00	1.30	0.99	1.10	0.90

Results *Fixed Budget Retrieval (k = 5)*

HotpotQA (Multi-hop reasoning)

- Hit@5 improves 0.10 → 0.95
→ ~9.5× improvement over Dense retrieval
- Multi-hop full evidence 0.10 → 0.48
→ Better coverage of multiple reasoning steps.

FEVER (Single-hop verification)

- Hit@5 remains ~1.00 across method
→ OT maintains accuracy while reducing redundant evidence.

SciFact (Multi-evidence scientific claims)

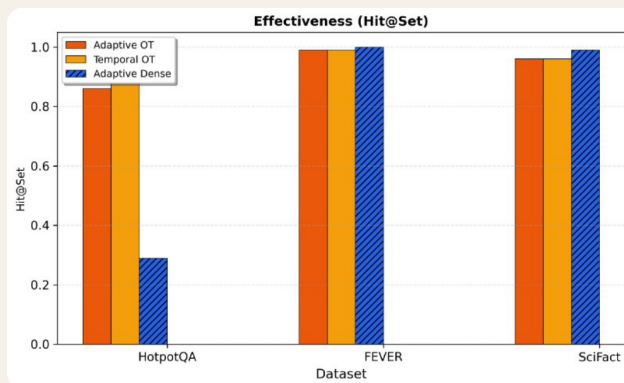
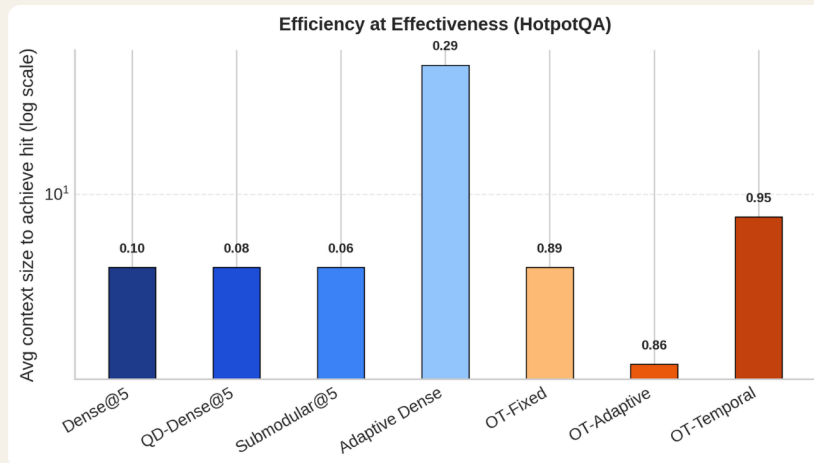
- Avg Gold evidence 1.12 → 2.00
→ ~78% improvement in evidence coverage
- Full evidence success 0.72 → 0.90.

Results

Adaptive retrieval effectiveness and efficiency with and without query decomposition.

Method	HotpotQA		FEVER		SciFact	
	Hit Ratio	Avg Size	Hit Ratio	Avg Size	Hit Ratio	Avg Size
Adaptive Dense	0.29	33.89	1.00	4.18	0.99	35.40
OT-Adaptive (no decomp)	0.86	2.00	0.99	2.00	0.96	2.00
OT-Adaptive (decomp)	0.85	3.39	0.90	2.18	0.88	2.17
OT-Temporal (no decomp)	0.94	6.00	1.00	6.00	0.98	6.00
OT-Temporal (decomp)	0.95	8.05	0.91	6.20	0.94	6.32

Efficiency



Effectiveness (Hit@Set)

- On HotpotQA, OT-Adaptive achieves 0.86 vs 0.29 (Adaptive Dense)
→ ~197% higher effectiveness
- On SciFact and FEVER, performance remains near 1.0.

Efficiency (Context Size)

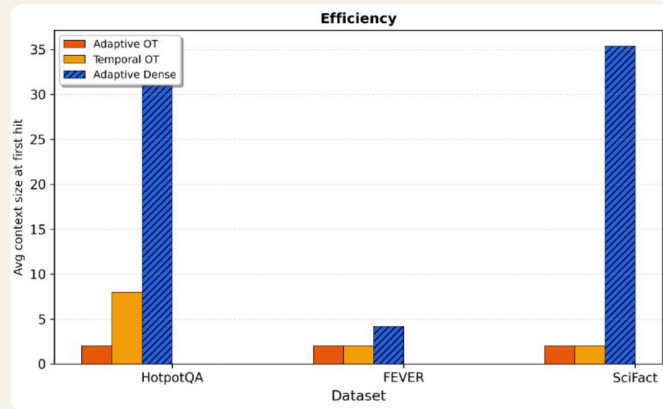
On *HotpotQA*:

- Adaptive Dense uses 33.89 sentences
- OT-Adaptive uses 3.39 sentences

→ ~90% reduction in context size

On *SciFact*:

→ ~94% reduction in required context



Coverage

Full evidence coverage across datasets.

Method	HotpotQA (MH-Full)	SciFact (ME-Full)
Best Dense@5	0.11	0.88
OT-Fixed@5	0.26	0.90
OT-Adaptive	0.27	0.92
OT-Temporal	0.35	0.95

Assumptions

Assumptions

Shared Semantic Space

- Query and candidate embeddings lie in the same vector space and are comparable.

Semantic Smoothness

- Cosine similarity reflects partial semantic overlap between texts.

Candidate Pool Contains Relevant Evidence

- Gold evidence exists within the provided candidate set.

Assumptions

Greedy Selection Approximates Optimal Coverage

- Marginal OT cost reduction reflects diminishing semantic gain.

Coverage Saturation Indicates Sufficiency

- When OT marginal gain becomes negligible, semantic coverage is considered sufficient.

Sentence-Level Atomicity

- Each sentence is treated as an independent evidence unit.

Work Breakdown

Phase	Nia	Richard	Sabharish	Sangeetha
Phase 1 – Research (Jun–Jul)	RL mechanisms LangGraph exploration	RL fundamentals Project concept study	RL on Knowledge Graphs Concept research Architecture design	Knowledge Graph basics Initial research
Phase 2 – KG + RL Development (Aug–Oct)	KG construction from own data Baseline evaluation (MRR)	RL partial reward handling Policy learning implementation Relation extraction (OpenAlex)	Reward design for RL RL training parameters Partial reward strategies	Dynamic KG building OpenAlex preprocessing
Phase 3 – Retrieval Reformulation (Oct–Dec)	SciFact candidate pool Dataset preprocessing	FEVER candidate pool Dense retrieval baseline evaluation	OT-based retrieval integration Temporal OT extension	HotpotQA candidate pool Dataset preprocessing

Phase	Timeline	Responsibilities (All Members)
Phase 4 – Evaluation & Optimization	January – February	OT variant implementation Temporal OT refinement Baseline comparisons Performance evaluation
Phase 5 – Documentation & Final Submission	February – March	Paper writing Final report preparation Experimental validation Final system consolidation

Requirements

Hardware

- Quad-Core CPU
- Minimum 8 GB RAM
- Storage (1-2 GB recommended)
- GPU Not Required

Software

- OS: Linux/Windows/MacOS
- Programming Language: Python 3.11

Libraries & Frameworks

- PyTorch - embedding computation
- HuggingFace Transformers - frozen encoder (BGE-base-en-v1.5)
- FAISS - dense candidate shortlisting
- NumPy / SciPy - numerical computation
- POT / custom Sinkhorn implementation
- Scikit-learn - evaluation utilities

Datasets

HotpotQA - Multi-Hop

- Supporting facts require two or more evidences
- Designed for multi-hop reasoning
- Each question is linked to supporting Wikipedia paragraphs
- Used to evaluate multi-hop evidence coverage

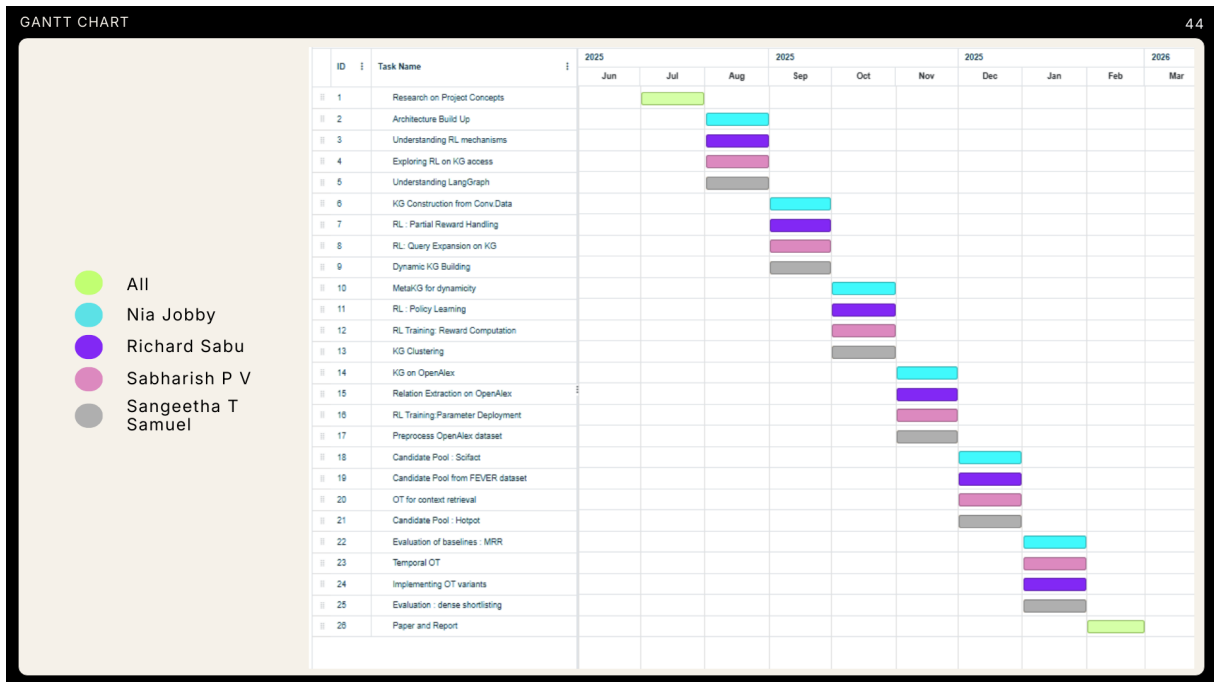
FEVER - Atomic Fact Verification

- Labels: Supported / Refuted / Not Enough Info
- Evidence retrieved from Wikipedia
- Mostly single-evidence dominant, but some multi-evidence cases

SciFact - Multi-Evidence

- Scientific Claim Verification
- Evidence drawn from scientific abstracts
- Many claims require multiple evidence sentences
- Reasoning-intensive dataset

Gantt Chart



Risks and Challenges

- High Computational Cost
Sinkhorn OT increases runtime compared to simple cosine ranking.
- Dependence on Dense Shortlisting
If relevant evidence is not in Top-K, OT cannot recover it.
- Early Stopping in Adaptive OT
Highly dominant evidence may trigger premature saturation.
- Scalability to Large Corpora
Memory and computation grow with candidate pool size.
- Multi-hop Reasoning Complexity
Sequential evidence aggregation adds overhead.

Conclusion

- Effective retrieval for reasoning tasks requires more than simply ranking relevant sentences, as traditional methods may not capture the full context needed for complex reasoning.
- By modelling semantic coverage using Optimal Transport, we were able to construct compact and meaningful context sets that represent different aspects of the query.
- These context sets better support multi-hop and evidence-based reasoning by allowing systems to connect multiple pieces of relevant information.
- Overall, the results show that reframing retrieval as a set-level optimization problem leads to clearer and more reliable context selection for downstream systems.

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Thank You!

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2. Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3. Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4. Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8. Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language, and learning differences.

PO10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11. Life-Long Learning: Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES (CO)

SNO	Description	Bloom's Taxonomy Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant applications.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	Level	Justification
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.

Mapping	Level	Justification
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.

Mapping	Level	Justification
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.